

“Padel analyzer” (sports advice app)



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INTRODUCTION: AI Coach / "Padel Analyzer"

We're developing a **mobile app** that will change the way you improve your athletic technique. Essentially, it's a **personal trainer powered by Artificial Intelligence (AI)**, ready to analyze and refine your every move.

To begin, we focused on **padel**. Why? It's a strategic game. The movements with the racket and the body position while playing are **"easy for AI to read."** This allows us to create a **coaching model that is not only accurate** but also scalable from the outset, giving you high-quality *feedback* immediately.

This *app* is designed for both individuals who train independently and coaches, as we believe it can be used as a tool to provide a clearer explanation of how to improve your techniques. However, it's primarily for people who train on their own or who can't afford a coach. We'll be your **"pocket coach."** The AI will give you **clear tips and visual examples** so you can try, correct, and level up at your own pace.

And this is just the beginning. Everything we learn with padel will be the **technological foundation** for the future. Analyzing racket movements will open the door for us to quickly integrate other similar sports, such as tennis or table *tennis*, which is why we believe starting with padel is a way to make our app easier to scale. Our vision is to turn this *app* into a **platform for athletic improvement** in many different disciplines.

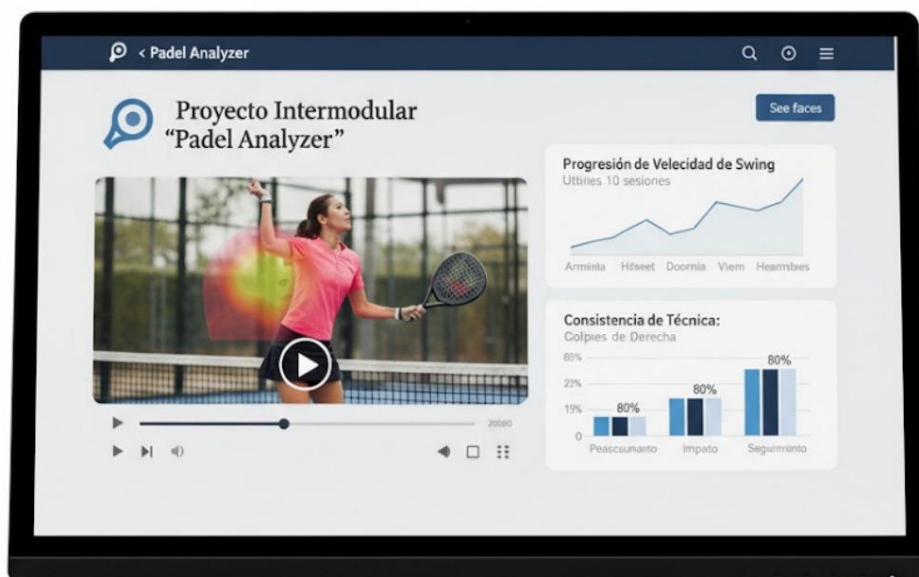


Figure 1: WEB AI Padel Analyzer



Application lifecycle

1. Project Start

- Definition of idea

The main idea of the project is an application that assesses your level and helps you improve through the use of AI, in short.

That would be the main idea, but it's not as simple as it seems, since it also tells you what you're doing wrong, how to fix it, and teaches you both text and videos to do so. It shows you how you're doing it wrong compared to how it should be done, and it also saves your improvement progress, tells you your level, and recommends movements to learn based on your characteristics.

It offers charts and graphs showing things like your consistency during the match, when you played best, heat maps indicating your movement patterns, which movements you repeat most often, and more. It displays a wealth of information and data from the match, showing some default settings, but you can always request more since it records everything during the game.

Even though it's been said that it analyzes you, it can also analyze your opponent, either in real time or by uploading videos. It will provide you with default data, just like it would for you, and give you a list of your opponent's strengths and weaknesses. This will show you how to counter their strengths and exploit their weaknesses to win your match. It will also take into account your movements and how to adapt them to achieve the best possible result.

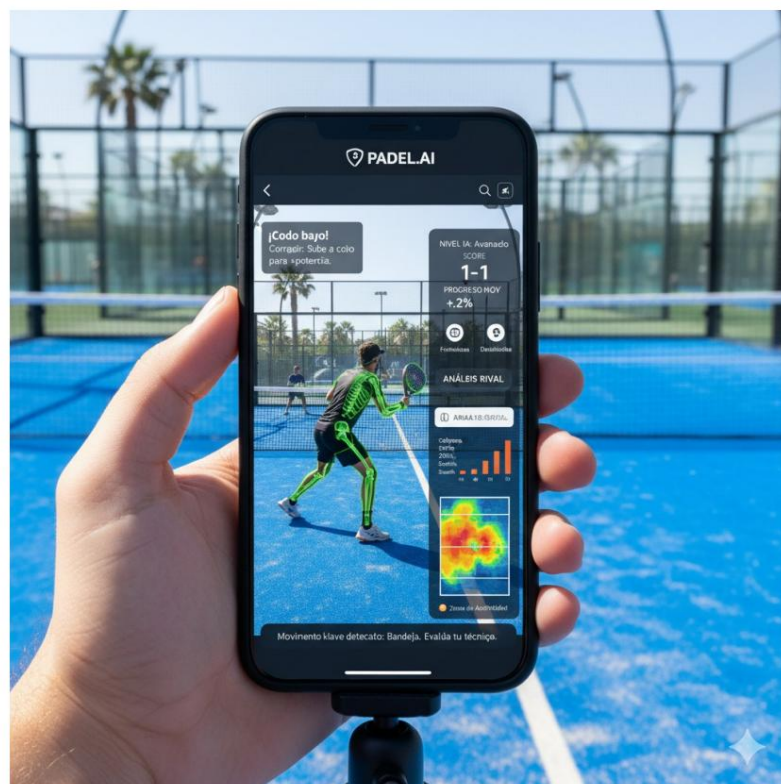


Figure 2: Example of use



Stakeholders

Stakeholders are companies or users themselves who seek to participate in the project or use it for their own benefit. For our project, I will divide them into four:

- **Padel equipment companies:**

- These companies could be from **rackets, clothing or footwear**, some of these are Bullpadel, Nox, Wilson, They could look for ways to use our products, whether it's padel rackets or clothing and footwear, in our educational videos.

We would be **showcasing their quality** and **advertising them**.



Figure 3: Stakeholders

They could also use our application to see how players use their rackets and improve them, or even use the application data **to create more ergonomic and specific designs** for each player, thus creating a **personalized**, high-quality product and raising their brand's profile. This applies not only to rackets but also to using heat maps to see the wear and tear on shoes and even clothing, to analyze which materials and designs are best, just as with rackets.

- **Sports clubs and academies:**

- Padel **clubs and court rental venues** could use our product on their courts by installing cameras and being able to provide a **coaching service** for customers, either free of charge or with a premium fee.

- For **academies and coaches**, it would be good to use them as in the previous case, but it would also support and help human coaches to see if they are doing it right.

- Some padel court booking management software might want us to integrate it into our app so that we can see their analysis of all the courts and book everything that interests you about padel in one app.

- **Media and content creation, ... :**

Streaming platforms, streamers, and padel **influencers** could use it to analyze **live matches** with more data and greater accuracy. This would also be ideal for creating newspaper articles and digital news stories.



- **Sponsors:**

ÿ Some brands might be interested, for example, in the event we will be holding (we will see it later), their drink could be related to padel, expanding their audience and associating themselves with the fact that it is useful for the sport.

Project feasibility

The project's viability is divided into three points: how feasible it is to be developed, sold, and maintained:

- The most complex aspect of the project is the development of **Artificial Intelligence** for movement evaluation. This requires significant work to create a model that can identify specific errors and compare the player's execution to the correct form. This phase requires an initial test to verify its functionality on mobile devices. However, the structure for **storing and classifying educational videos** by level and movement is already created, thus allowing for content scalability. Once AI is developed, integrating the analyzed data, such as **heat maps** and **statistical tables**, is a standard function.
- Respect for **sales and commercial aspects** , since padel is constantly growing, ensures the demand for our product because it is something new, innovative and most importantly, **unique**.
- Creating the **library of educational videos**, organized by levels and movements, is a resource-intensive process (time, recording equipment, and model players), but it's a **one-time investment** that ensures the quality of the core product: teaching through videos. Regarding partnerships, collaborations with **sports clubs** and **padel equipment companies** facilitate market entry and offer the possibility of integration into premium services. The product is digital, which implies **high global scalability** once the AI is developed, minimizing long-term operating costs.



Project Charter

Project Name: Padel Analyzer & AI Coach

Date: 27/09/25

Project Director: Rodrigo Viso Cain

Project justification:

Padel is a sport experiencing exponential growth globally. Players of all levels are looking for objective and accessible tools to improve their technique and performance. The cost of a qualified personal coach is a barrier to continuous improvement.

This project is justified by the need to digitize personalized training, offering an elite diagnostic tool (based on AI) that is scalable, affordable, and capable of providing immediate feedback, advanced statistics (heat maps), and personalized improvement plans. This positions PADEL.AI as the leading solution for technical improvement in padel.

Goals:

Develop and launch a functional mobile application that uses Artificial Intelligence to assess the user's gaming level and offer a personalized improvement plan based on the identification of technical errors.

Achieve a motion error diagnosis accuracy of over 90% in the beta testing phase.

Reach 10,000 premium subscription users within the first six months after launch.



2. Planning

Project Objectives

-Main Objective: To offer instant and accurate technical feedback through automatic analysis of the user's posture, movements, strokes and displacements during matches or training.

-Technological Objective: To integrate AI capable of identifying with a high accuracy of 90% the errors in the technique of the blows, in the positioning and in the movements of the users.

-Functional Objective: To launch a version that allows recording, analyzing the recording, and storing this information so that it can be used and displayed in statistics and analyses.

-Quality Objective: Implement a high-quality system that provides feedback and useful advice to users, along with objectives to be met gradually so that they can progressively improve their level of play.

-Performance Objectives: Ensure the processing of multiple long-duration videos without failures or performance degradation, optimize cloud storage, and maintain a stable application without crashes or serious errors.



Figure 4: (Planning) Movement Analysis



Tasks required to complete the project

To complete the project, we divided the tasks into several phases:

Phase 1: Research and data acquisition for the AI.

-Definition of Metrics: Establishing what errors, hits and positions are analyzed and how they are analyzed.

-Data Collection: Record or acquire thousands of hours of padel videos labeled to indicate correct and incorrect shots, positioning errors, etc.



-Data Preprocessing: Normalizing, cleaning, and labeling the dataset for AI training.

Figure 5: task planning

Phase 2: AI Model Development

-Model Training: Train the model with collected data.

-Optimization and testing: Adjust the model to improve accuracy and achieve that 90% accuracy and reduce processing time.

Phase 3: Development of the mobile application and its interface

-Design: Design the user interface for a simple experience of recording, uploading and viewing results.

-Development: Create the cloud infrastructure to host and run the AI model and program the application with recording and analytics visualization functions.

Phase 4: Integration and testing

-AI Integration: Connecting the mobile application with the execution of the AI model.

-Development of tips: Schedule the generation of personalized tips based on the errors detected.

-Testing: Conduct internal tests and then with a small group of users (beta-testers) to identify bugs, analysis failures and verify that there is 90% accuracy in the AI.



Resources needed

-Human Resources:

- 1 AI Engineer responsible for designing, training, optimizing the AI model and achieving 90% accuracy.
- 1 Developer (Cloud + Backend) to build the cloud infrastructure to manage video uploads, run the AI model, and store the results.
- 1 Mobile application developer (multiplatform) responsible for video recording, efficient cloud uploading, notification management, and interactive visualization of analysis results.

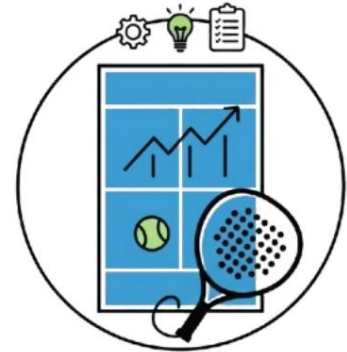


Figure 6: planning-resources

- 1 Padel Expert/Coach to define the analysis metrics and validate the accuracy of the AI.

-Technical Resources:

- Cloud servers containing GPUs for intensive AI processing.
- Development software (IDEs and collaboration platforms for managing code and versions).
- Databases for high-quality padel videos.
- Data labeling and annotation tools: specific software so that the padel expert/coach can mark the locations of errors, positioning, technique in the strokes, etc.

-Financial resources:

- Budget for team salaries that must ensure the retention of key personnel.
- Cloud infrastructure costs, including video storage, GPU processing power for AI, and data traffic
- Marketing budget, which should cover brand development, content creation, and digital advertising to boost app downloads.



Schedule

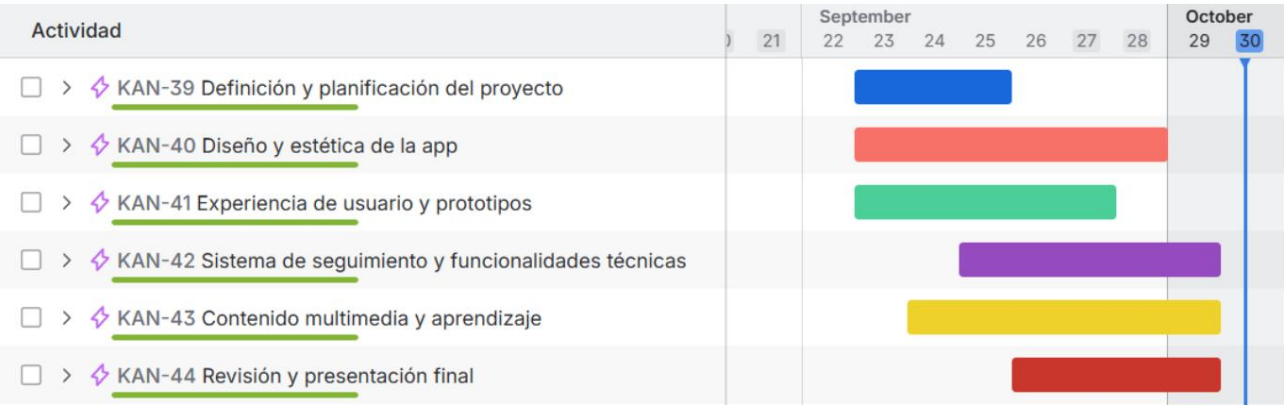


Figure 7: planning and execution

The schedule shows the activities to be carried out for our application, which are developed during the month of September:

Project definition and planning: It begins on September 23rd and ends around the 25th. September.

App design and aesthetics: It extends from around September 23rd to around September 29th October.

User experience and prototypes: Starts around September 25th and ends before October 28th.

Tracking system and technical functionalities: It starts around September 26th and It extends beyond October 30.

Multimedia content and learning: It begins around September 27th and extends beyond October 30.

Final review and presentation: Begins around October 28th and runs until just before October 30th or a little later.



Risk analysis

Possible risks and their possible solutions:

-Low AI accuracy: Diversify the data source, hire an additional expert, and initially focus only on the most common errors, expanding later if you cannot achieve 90% accuracy.

-High cloud processing costs: Optimize the model to reduce GPU usage, try to reach an agreement with the cloud provider, and initially limit the maximum duration of the video analysis and extend it later.

-Legal/privacy issues: Hire legal advice to ensure compliance with data protection regulations and establish a clear privacy policy.

-Low adoption and poor user experience: Conduct surveys of padel players before the beta launch to validate the need, perform early usability tests and validate the content through the padel expert/coach.

-Schedule Delay: Allocate more resources to the corresponding phases and use agile methodologies with weekly deliveries to detect where the deviation is occurring. time.



3. EXECUTION

- Technical development of the application

* A video analysis system using artificial intelligence will be implemented to detect and evaluate the player's movements.

We know of several artificial intelligence applications that can detect and evaluate a player's movements.

We will use applications such as:

OpenPose: Allows us to estimate the player's pose (captures movement from the mobile phone's camera)

<https://openposeai.com/>

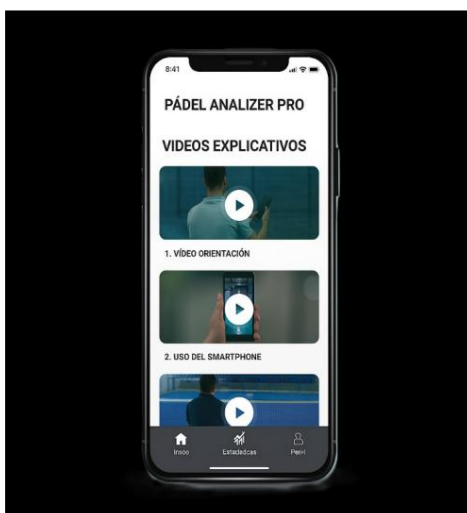
Arthlete: Allows us to correct and analyze player movements in real time <https://www.athlete.fit/>

And more...

With these applications we can improve our project

* Algorithms will be designed to calculate some key indicators, such as:

- Reaction speed from when the opponent hits the ball until the player responds (reaction time)
- Time from when the player moves towards the ball
- Most frequent errors made by the player in the execution of the strokes
- Percentage of hits on offensive and defensive shots



Our system will present the user with visual and statistical reports on their performance. These will include specific

recommendations for improvement, such as improving foot positioning, reacting faster to the ball, or perfecting the technique of a shot.

* A library of explanatory videos created by padel experts will be integrated.

The videos will be categorized and automatically suggested based on weaknesses detected by our application.

Figure 8: Design execution



- Team management and resource coordination

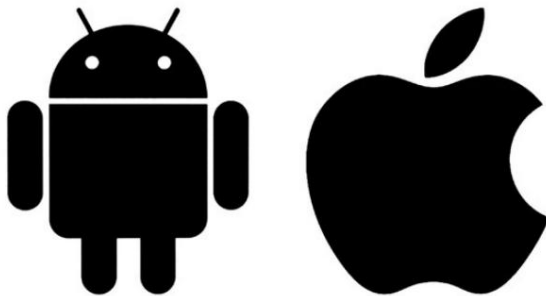
The organization of our team is fundamental to ensuring that each part progresses in a way that is effective. balanced

Distribution of roles:

- Developers: They program the application, implement the AI algorithms, and test its performance
- UX/UI Designers (digital products): Building intuitive and attractive interfaces for the user
- Padel experts: Validation of technical parameters, definition of quality criteria in the analysis of the movements
- Testers: Execution of functional, usability, and performance tests on different devices

Technological resources:

Use of cross-platform development frameworks, in which compatibility can be ensured on iOS and Android



-They facilitate the creation of intuitive and attractive interfaces, with fluid animations, which can be important for displaying motion analysis more clearly.

-They can be easily updated with new features (incorporating advanced statistics, a comparison module with professional players...)

Figure 9: Resource execution

Servers to efficiently process video data:

- They will be able to store videos and training results in the cloud, so that the user can check your history at any time
- Metrics can be shared and compared with other players (rankings, progress, etc.)

Test devices:

The application will need to be tested on different mobile phone models to guarantee the compatibility and optimal performance

- Low-end: Check that the application works acceptably on devices with limited memory and processors
- Mid-range: Validate that most users have a smooth experience
- High-end: Take advantage of all available power (high-resolution videos, etc.)

Different usage conditions will also be simulated.

- Recordings with bright light and low light
- Scenarios with different backgrounds, so we can ensure that the AI does not fail in detecting poses



-Variable distances from the mobile device to the player (closer, farther)

- Monitoring and control of progress

Supervision will ensure that the project does not deviate from the initial objectives.

Progress monitoring: - Regular

meetings will be held with team members to review the progress of each area. These meetings will serve to prioritize activities and make quick decisions to avoid delays in the schedule.



Tools like Jira will be used to visualize the status of each task. With this tool, each team member can check their responsibilities and report issues immediately.

Figure 10: Execution, planning, and control –

Collaboration, communication, and team member satisfaction will be evaluated. This aims to maintain a motivating and efficient work dynamic.

Quality control: -Tests

will be carried out to verify that the artificial intelligence vision algorithms correctly identify the player's movements, body position and execution of the strokes, which will serve to measure the accuracy of the system.

-Comparative tests will be conducted between the application's automated analysis and the evaluation performed by professional coaches. -Each version of the application will be tested, feedback will be gathered, and improvements will be implemented before release.
advance

Corrective measures:

Corrective measures will be essential to ensure that the project stays on track in the face of possible deviations in time, cost, or quality.

The process will begin with the early detection of incidents, which will be achieved through the constant monitoring of performance indicators; they can also be identified through the user testing phase.

Once a deviation has been identified, a root cause analysis will be performed to determine the origin of the problem.

This will allow us to define the most appropriate action, thus preventing the solution from being too simple.



The corrective measures adopted will be documented and communicated to all interested parties, so we can ensure that there is a clear record of the adjustments made and their impact on the overall planning.

This will also ensure the continuous improvement of the development process

- Communication with stakeholders

Regular meetings:

-Weekly or bi-weekly meetings will be held with stakeholders, the development team, padel experts, investors...

These meetings will allow us to present progress, validate functionalities, and gather suggestions that will help us adjust the product in real time.



Figure 11: Meeting execution

Status reports:

-Status reports will be prepared at a defined frequency (weekly or bi-weekly) and will include key progress indicators, technical metrics, and test results.

Some reports will be distributed through digital channels (email) which will ensure their availability

Bidirectional feedback channels:

-Two-way feedback channels will be implemented, allowing test users and sports experts to provide comments, opinions, point out problems, and propose improvements



01/10/2025

Padel Analyzer

30/09/25

This feedback will be categorized for integration into the development cycle, ensuring that the application evolves according to the needs of the end user.

Transparency protocol for risk management: - In the event of

identified delays, technical difficulties, etc., stakeholders will be notified immediately, along with the defined contingency measures. This will help maintain confidence in the project and ensure effective collaboration throughout.
moment



4. Monitoring and control

- Monitoring and control of our application will be divided into three key points: •

Project performance will be measured against our goals. • Adjustments will be made to replan tasks or reallocate resources. • Maintenance costs and application quality will be monitored.

- Project measurement / Are we achieving our goals?

The project's success will be measured through the progressive achievement of goals. These have been defined based on the **functionality, accuracy, and scalability** of the Artificial Intelligence (AI) solution in the application.

Short-Term Goals (Minimum Viable Product - MVP)

1. Development of a **frame-by-frame analysis** functionality that **correctly** analyzes **short videos** to provide a decent analysis of your technique, your strengths, and how to improve it.
2. That the **application** can already **be used** , even in its **simplest version**, such as
For example, the **graphs** show your consistency and heat zones.
3. And that the **AI** gives **good advice** on how **to improve** not only techniques but also training methods, such as progressions to improve your performance during matches.

Medium-Term Goals (First Complete Version - V1.0)

1. **The launch of version 1.0** of the app will simply be a larger-scale test of the application's **basic functionalities** .
2. If possible, we would like to get **feedback from users** with the **paid plan** of the application to know how accurate our AI is when analyzing the videos and whether they consider the advice to be good.
3. Have a **plan for future updates**, such as the **real-time coach**, which in
The principle is like an extension of the short video analysis principle.

Long-Term Goals (Expansion and Added Value)

1. Launch of the **"AI Personal Trainer"** functionality that offers correction in real time or during gameplay.
2. **Addition of 3 new sports** to the catalog with adapted AI analysis functionality.
3. **Sustainability and Quality Control.**



- Task and resource adjustments

The project will take into account that sometimes things don't go as planned or we can't maintain a certain level of quality because we lack the necessary resources; examples of this can be

1. The AI does not reach the minimum accuracy threshold (e.g., less than 75% in the PMV).

- **Absolute Prioritization of the AI Core:** The scope of all non-critical tasks for the MVP is **temporarily stopped** or **reduced**, such as the development of the real-time assistant or some other high-quality graphic options that are not part of the MVP.

2. We don't have videos to analyze = programmers ready (lack of "video" resources)

- For **Artificial Intelligence (AI)** to analyze tennis, golf, or basketball, it needs to watch thousands of **example videos**. This is like giving textbooks to a student before an exam.

- **Temporary Hiring:** We hire a **freelancer** (an external expert) or an agency to help us gather many video examples from different sports so we can give the AI a break and have several programmers fix app bugs and move forward in other areas.

3. The application works correctly but is not well known

- To start seeing **benefits**, we need **to raise awareness of our app**. If we're already using social media, we can change the way we make our videos, or talk to a **fitness influencer** who might be interested in the app and want to help us by using it in their videos and teaching all kinds of users—trainers, average players, etc.—how to use it.

Padel Influencers

Here are Top 35 Padel Influencers to Follow on Instagram in 2025

1. Ale Galan



Bio Professional Padel Player 🏸 @adidaspadel ****@380amk.com
Instagram Handle @alegalan96
Instagram Followers 680.3K · **Type** Macro · **Gender** Male

✉ Get Email Contact

Figure 12: (Follow-up) Padel influencer



- Control of Maintenance Costs and Application Quality

This point is fundamental to guaranteeing the project's **sustainability** in the medium and long term, a Once the goals of the **PMV** and **V1.0** have been achieved, control is focused on two pillars:

Quality Assurance (QA) and **Operational Cost Optimization**.

A. Quality Control and Assurance (QA)

- Ongoing mechanisms are established to ensure that the application maintains a high **level of quality** and **reliability**, which directly impacts **user satisfaction** (and therefore, the payment plan duration).

- **Stability and Errors:** A **record of critical bugs** is maintained and the **time to resolution** is measured.

Activity - Objective: Resolution of critical *bugs* in less than X hours. Failure to comply will affect directly **Sustainability and Quality Control** (long-term goal), noting that the Maintenance is not "decent" and requires reallocation of resources towards stabilization.

- **User Experience:** **Feedback from users with a paid plan** is used (media goal) (term) and satisfaction surveys to evaluate **usability** and the **quality of the advice** generated. If the application works correctly but users do not understand it (similar to the Point 3 , "is not well known"), tasks are readjusted prioritizing the clarity of the graphics and the interface on the development of new *functionalities*.

B. Control and Optimization of Operational Costs

- Developing an application with **Artificial Intelligence** involves high operating costs (servers, storage, computing). It is vital to establish strict control to ensure the **financial viability** of the project.

- It is ensured that the **temporary hiring of external experts** (e.g., for **data collection**) (**video** according to point 2 of the image) is **strictly limited by time and deliverables**.

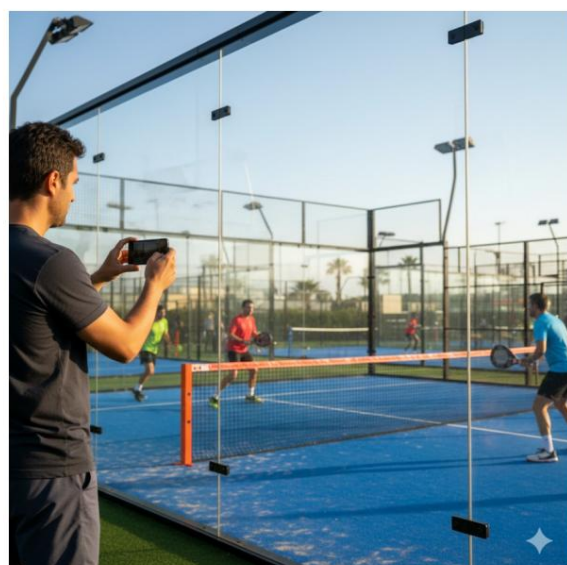


Figure 13: (Follow-up) Recording for AI training



5. Project Closure

Verification

In this phase, we focus on ensuring that everything promised has been delivered. First of all, They will perform **quality tests** which the **AI Engineer** and the **Padel Expert/Trainer** will carry out **They will test the AI**, running load tests **simulating the simultaneous upload of** Several videos will be shown and **various analysis simulations will be reviewed**, analyzing the feedback and the Tips will be generated and **verified by the Padel Expert/Coach**. Based on what In those tests, we will verify that the following is true:

-Mobile application: This must contain a complete functionality for recording, uploading and viewing analysis with a high failure stability of less than 2%, this will be verified by the cross-platform application developer.

-AI Model: This must have A high degree of accuracy in identifying key errors during analysis, exceeding 80%, must be verified by the AI Engineer and the padel expert/coach who will provide information to determine if the AI is functioning correctly.



Figure 14: Closure verification

- Advice and feedback system: This should consist of a personalized generation of advice and feedback, approved by the padel expert/coach for each error detected in each analysis; this should be verified by the padel expert and the main project leaders.

-Cloud infrastructure: This must be a cloud structure, which has security mechanisms, such as data backups; the backend/cloud developer will be responsible for verifying this.



Project documentation and archiving

In this phase we collect and organize all the documents generated during the project:

-Code documentation: Archive all code repositories in secure storage, including API documentation and deployment procedures. A freeze-frame of the final application version will also be created and stored in secure repositories, including automated deployment scripts.

-Cloud architecture diagram: This will be a graphical document that details exactly how the cloud service is configured, including GPU instances, databases, storage services, etc.

-AI Model Documentation:

It should be a detailed document (Figure 15: *closing documentation*) about the dataset used for development, the architecture, the results of the accuracy tests, and the instructions for training and retraining the AI Model.

-Operation manuals: Create operation manuals for the team responsible for maintaining the application in good condition, detailing the management of the cloud infrastructure, cost monitoring, and procedures for solving common problems in the application.

-Design archiving: Save the application design files, logo, marketing resources, brand manuals, etc. All with a clear description of what each one is and what it does.





EVALUATION AND ARCHIVING OF LESSONS LEARNED

Final project evaluation (Post-Mortem)

A formal evaluation meeting is held with the core team to conduct an analysis of the project's performance.

Performance Analysis:

Time and budget:

- We analyze whether the project was completed on time and is within budget.
- We analyzed whether there were any deviations
- What are the causes

Quality of deliverables:

- We analyze whether the AI model and application meet the defined quality standards
- We analyzed the critical feedback received during the beta version

Planning effectiveness: _____ of _____ the _____

- We analyzed whether the risk analyses and resource estimation were effective

Evaluation of the Team and Processes:

Efficiency of the tools used: _____

- We evaluated the efficiency of the collaboration tools and communication between the technical team and the padel experts

Frequency and quality of communication: _____

- An assessment will be made to determine if the meetings were productive, if everything was clear, and if fluid communication was maintained between the technical team and the padel experts.

Conflict resolution: _____

- A record will be made of any conflicts that arose and how they were handled.
Whether they were resolved constructively and whether there were clear protocols for decision-making

Lessons Learned and Future Recommendations



Area	Lesson learned	Future recommendations
AI and Data	Manual labeling was slower than expected to be labeled from the beginning This was due to the variability of the recording angle guides.	We must prioritize tools over what we Invest in standardized user-friendly
Cloud technology: The cost of GPU processing	underestimated by 15% in the concept phase of finalizing the budget	We would conduct a test that was processing with a real-load test before
Processes	The integration of the opinion of the to the AI model was faster, where experts are slow, requiring them to validate small batches of data, thus allowing for incremental adjustments to the model.	Creating a pipeline of expert padel opinions

Final File and Transition

Centralized Archive:

- All documents, evaluation reports and lessons learned will be archived in a central repository.

It will be accessible to project management

This will facilitate future consultations, allow us to replicate best practices, and prevent the loss of valuable information.

Resource Release:

We will officially end the team's participation in the project:

- Closure of contracts or temporary assignments
- Reassignment of staff to new projects or functions
- Transfer of responsibilities

Closing Celebration:

It will be a moment to recognize the effort of the entire team, we will thank them for their work and celebrate the achievements reached.

This will reinforce our sense of accomplishment, improve the work environment, and motivate the team for future challenges.



Tasks from day 1 to 7

- DAY 1

1. Vision and Value Proposition

Vision: To create the ultimate application, powered by **Artificial Intelligence** and **Video Analysis**, so that padel players of any level can diagnose, correct and **improve their technique and performance** autonomously.

Main Objective: To offer **instant and accurate technical feedback** through automatic analysis of the user's posture, movements, strokes and displacements during matches or training.

Differentiation: Combining in-depth technical analysis with **social motivation** and **gamification** to encourage continuous improvement and friendly competition.



Figure 16: (DIA1) Vision and value proposition



2. APPLICATION SKETCH:



Figure 17: (DIA1) Sketch

2.1. First section Start recording:

- **Camera Button (Large center circle):** Pressing this button starts the process of recording a new padel session. This may activate the phone's camera or connect to an external camera to capture the game and for post-game analysis.
- **Tripod Icon (Right Corner):** Suggests that the application is designed to be used with a fixed setup (a tripod) for the best angle and analysis.
- **Function: To capture and analyze real-time or delayed data** from a match or training session.

2.2. Second section:

- **Title and description ("Session Library"):** Indicates that all completed recordings and analysis reports are stored here.
- **Right arrow (→):** Typical of an element that takes you to a new screen or list.
- **Function: View, review, and compare** previous sessions. The user can see metrics, tagged videos, and heatmaps for each match.
past.

2.3. Third section (This section summarizes the metrics from the previous session; the actual metrics are found after entering the "My Analysis" section, which displays the metrics for the selected video).

- Bar Chart
- Track Heat Map (RANKING - right):
- Function: To provide a quick and detailed overview of performance



3. Research of competitors and similar apps

Next, we will research the sector we are entering by examining competing applications. This will help us identify features that may be in demand and that are currently unavailable in the padel app market. We will also determine which features currently offered by competitors could be improved.

3.1: First type of competition

Within the padel app sector, the first type of apps we find are those for **booking courts** or **finding someone to play with** near you. They also offer in-person private lessons in some cases.

cases.

One example is the **Playtomic app**, which allows us to book matches, lessons with instructors, record our scores, and track our progress. It's geared towards padel, tennis, and other racket sports.

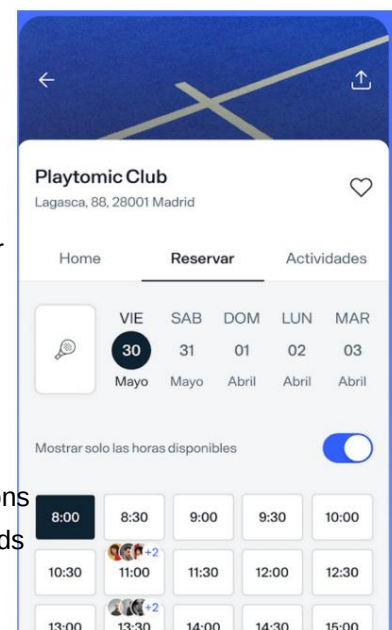


Figure 18:
(DAY1) Competition

The problem with these apps is that when it comes to learning on your own counts or improving in padel, **it's all in person, it's an extra cost monetary and temporary** that many people cannot afford.

And performance is simply measured in general by match scores; **there is no exhaustive analysis of the player's** specific plays or techniques.

In the future we could consider adding the option to reserve courts or play in person with other people, but since there is already a lot of competition regarding this function.



Figure 19:
(DAY1) Competition



3.2: Second type of competition

Later, we will realize that the other type of Applications related to padel are those related to the **learning** this sport. These are our biggest competitions. direct, since it is more focused on improving remote padel and not both in-person and the first type of applications.

An example of this type of application is **Prodigy Padel Academy**. This application focuses on teaching through structured videos divided into different sections and accompanied by text for each video.

After each video, the app suggests exercises. illustrated with drawings or animations, so you can practice it taught previously.

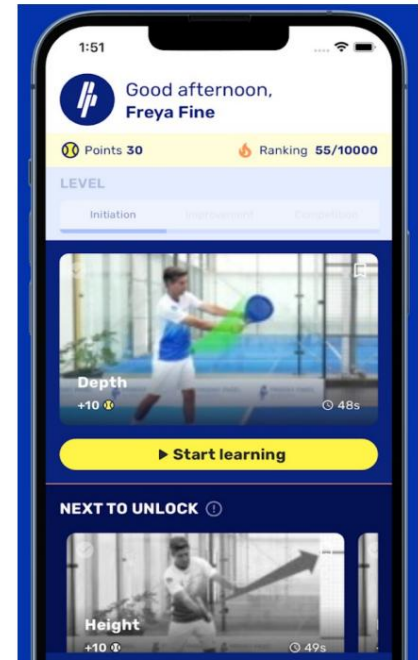


Figure 20:
(DAY1)Competition

The main problem with this type of application is its **lack of adaptability**. and **user customization**. These are predetermined courses for any user.



Figure 21: (DAY1)Competition

Which do not adapt to the needs and things that it has to
Improving the padel player is **a waste of the user's time**.
Making him repeat or pay for content that is unnecessary for him.

We will focus on **showing only the videos** and on **giving tips** for improving the specific things the user needs
after playing. of analyze his shape of

This will allow the player to save time and improve faster at padel.
focus only on the mistakes he makes or things he has to
Improve. And customized for each type of play style and style.
user level.



4. BRAINSTORMING OF NAMES

4.1. *PadelAI Coach*

This name is direct and leaves no doubt about the value proposition. It uses the term "**AI**". (**Artificial Intelligence**) directly to position itself as a technological solution and the term "**Coach**" to communicate that it offers advice and improvement.

4.2. *PadelMotion Pro*

This name is **dynamic** and action-oriented. "**Motion**" refers to **frame-by-frame motion** analysis (one of your key goals), and "**Pro**" suggests an **advanced, high-precision level of analysis**.

4.3. *Vision TrackAI*

A name that evokes **vision** and **tracking technology**. "**Vision**" and "**Track**" are common terms in camera-based sports analysis, and adding "**AI**" at the end communicates that the accuracy and corrections are the result of an intelligent engine. It's a name that sounds robust and professional.

4.4. *Padel Analyzer Pro*

Keeping the "**Analyzer Pro**" is a safe choice. It's **clear and descriptive**. It tells the user exactly what they're getting: a **detailed analysis** of their performance. Although it doesn't include "AI," the "**Pro**" suffix implies a professional-grade tool that goes beyond simple recording.

4.5. *Enhance Padel AI*

This name uses a strong verb ("**Enhance**" or "Improve") to emphasize the **benefit** the user will gain. It's less technical and more solution-oriented. The addition of "**Padel AI**" keeps it modern, making it clear that this improvement is achieved through artificial intelligence.



5. Role assignment

Clearly, this is one of the most important parts of the project because dividing the work clearly and systematically makes it much easier, more efficient, and promotes a better final result, since everyone has a clear understanding of their own objective and can focus on it; that's why it's one of the first tasks.

In this project, two main roles can be distinguished: the project leader/coordinator, Mario, and the rest of us, who are team members with the same level of responsibility but for different aspects. This means that each of us is in charge of a different part, making our own decisions individually, but always with Mario's approval, as he is the expert in the field (padel). He helps us adapt and understands the needs, tastes, and preferences of our padel client, from the average player to the professional.



Figure 22: (DAY 1) Padel training

The roles of each person in this project are explained below:

- **Mario** ÿ Project Coordinator / Leader.
 - Its function is to decide the direction of the application, what functions to implement, what aesthetics to give it to make it attractive to its audience, how it will be measured and on what parameters the AI will be based. This means that it will guide the direction of the project and adjust the important details of its operation.
- **Sergio** ÿ Responsible for design and tracking system.
 - Responsible for the application's visual identity, colors, logo, Also
Among its tasks is analyzing competitors and similar applications to ensure the proper execution of its other duties. In addition, the system records and saves player movements and statistics, ensuring data is stored correctly and that the application has an attractive and intuitive interface.
- **Rodrigo** ÿ Interface and web page designer.
 - Create the app and website designs, ensuring consistency and user-friendliness, including menus and different pages/sections. Test the interface navigation to ensure it suits the user.



- **Cristina** ÿ User experience and communication specialist.
 - It focuses on the player's understanding of the application, designs usage examples, Ensure the functions are intuitive and simple. Also, plan how the that all this information and functions be presented to the user, and finally also organizes communication with the public through social networks.
- **Antonio** ÿ In charge of promotion and multimedia content.
 - The work is based on preparing the dissemination of the project through different media and channels, organization of explainer videos, promotional videos, and of course what This entails the production, editing, and subsequent publication of the content adapted to the social network or if it is for the application and the target audience.

- DAY 2

1. Market Trends and Best Practices

The analysis of successful sports applications reveals essential keys to design and functionality:

Application Reference	Key to Success	Practice to Adopt
Tennis SwingVision Technical Analysis	Advanced with AI.	Integrating AI for detection and correction specific technical errors in padel (angle of the shovel, point of impact).
Nike Training Club Strong gamification and community.	weekly challenges	Implement rewards (achievements, medals), statistics to encourage competition.
Runkeeper	Clarity in metrics and interface. Easy-to-read graphics	Prioritize a visual and simple interface , with statistics not complex. Export to Spreadsheets

Conclusion: Our application should be a fusion of **powerful technical analysis**, **experience of intuitive user experience** and **social/competitive engagement**.



2. Video analysis section (initial sketches):



This screen is designed for the user to **review, study and understand** their detailed performance in a specific padel session.

2.1. Video Player (Main Video)

- **Function:** Display the **recording of the analyzed session**.
- **Utility:** It allows the user to see the plays in context. Crucially, this video likely has **labels or markers** that sync with the data below. For example, tapping on an anomaly in the graph would automatically jump the video to that moment.

of the play.

2.2. Impact Zones (Heat Map)

- **Function:** To visualize where the **contact with the ball** occurred on the paddle, or where **the player's impacts** occurred on the court (depending on the type of analysis).
- **Usefulness:** Identifies the areas of the paddle you hit to evaluate the consistency of your stroke. If it represents the court, it shows the most active areas or those with the greatest physical wear.

Figure 23: (DAY2) Session Analysis

2.3. Key Metrics

- **Reaction Time:**
 - **Function:** Measures the time it takes the player to **initiate movement** after that the opponent hits the ball.
 - **Usefulness:** It serves to evaluate **mental and physical agility** and anticipation.

**Impact Effectiveness:**

- **Function:** Shows the **percentage of** hits analyzed (i.e., hits) that bypassed the net and were effective, without being unforced errors).
- **Usefulness:** It offers a clear metric on the **reliability and consistency** of the game of user.

2.4. Evolution Chart

- **Function:** Shows the **trend of a key metric** throughout the session, in this case, the **"Average Speed"**.
- **Utility:** Allows the user to see if their performance (ball speed, intensity, etc.) **It either held steady or declined** as the game progressed.

2.5. "View Play Details" Button

- **Function:** It is the gateway to an **even more granular analysis**.
- **Usefulness:** Tapping this button will likely open a detailed, play-by-play list where the user can **filter** by shot type (smash, volley), result (winner, unforced error), or point in the match. This is essential for the **coach or advanced player** seeking the highest level of detail.



3. Research competitors and similar apps

3.1.: Third type of competition

Then we have a new and very small group of applications that are our most direct competition, which focus, using Artificial Intelligence, on scanning the playing style and analyzing the errors of the padel player or similar racket sports player.

We found four main applications: **GameCam (GameTraQ)**, **Aiball** and **SwingVision**.

3.2. Type of applications geared towards commercial use

On one hand, we have **GameCam (GameTraQ)** and **Aiball**, which are geared more towards commercial use. They utilize very high-resolution, high-quality cameras installed in sports centers. Clients can then use their app to view their performance during the match, along with relevant information.

Exactly.

The problem is that it doesn't help the user improve, and it requires a **very cumbersome and expensive installation** (you can see this to the right of the cameras **GameCam** uses). That's why it's a competitor, but not a major one, since it's not geared towards the same type of customer or needs. You can only see the performance, but there are no **recommendations for improvement** for the user.



Figure 24: (DIA2)
APPS

3.3. Type of applications geared towards personal use

Our most direct competitor would be **SwingVision**; it is the only application we have found that seeks to analyze the performance of users when playing (although in this case it is only with Tennis and PickleBall, although it could easily expand) using only the camera of their mobile phone.



4. USER FLOW

4.1. Registration / Login

- **What we're going to implement:** A system that allows users to **create an account** using their email or accounts from popular services (like Google or Apple). For **us**, it's vital that this process is very simple so that people don't abandon the application before even starting.

4.2. Camera Configuration

We need to make sure that the recorded video is of sufficient quality so that **our AI** can analyze it correctly.

- **What we will implement:** A clear **visual guide** within the application. **We will indicate** to the user the **optimal position** of the phone (for example, in the center or in a corner) so that the **AI** has a **complete view** of all the action on the court.

4.3. Record Match or Training

- **What we will implement:** A **large, intuitive button** to start and stop recording. Once recording is complete, the application should **automatically upload the video** to **our servers**. **We will display** a confirmation message to the user to let them know the analysis process has begun.

4.4. Automatic Analysis

This is the heart of the application, where **our Artificial Intelligence** works its magic.

- **What we will implement:** The **AI engine** we **have trained**. This engine will receive the video and, on our servers, analyze it **frame by frame**. The AI will identify **body position, striking angles**, and movement **consistency**. From this phase, **we will obtain** a list of technical data.

4.5. Report + Tips

- **What we will implement:** The **reporting interface**. **We will display consistency graphs** and hotspots **in** a highly visual way. Most importantly, **we will use AI** to generate **clear phrases** that identify the problem (e.g., "You need to improve your arm extension on the volley") and suggest **training methods** to correct it.



4.6. Access to Improvement Videos

- **What we will implement:** An **organized library** of **short training videos**. When **our report** identifies a specific error, **we will link** that error directly to the video demonstrating the correct technique or exercise. This way, the user has the help they need instantly.

5. Video Learning Planning

This task consists of planning what content we will include from the beginning in the application for learning and correction, so we will search for many movements and techniques to later select them once recorded and identify what level, type of movement, etc. it belongs to.

Not only that, we have to see where to record it, who will record it, how many people will appear in the videos and who they are, plan everything, the different types of videos if we want to use different techniques depending on the level, whether we group them or if we record everything and then directly relate the error or technique to be learned to the video according to the selected or perceived level, and not forget not only the level, if the goal is to help a professional, the language must also be adapted, etc., or if it is for a child.

After searching for a suitable location for the recording, I think it could be at [Pàdel Play Sant Vicent](#). Since it's a well-known and familiar place for the padel community here in Alicante, this could bring us closer to customers in the area and also allow us to expand our content, even hold some kind of promotional event, relying on them as a trusted venue, or use it to record content for social media, although we'll discuss this topic later.

Let's not forget that it has a contemporary aesthetic, which is why it benefits us over time.



Figure 25: Outdoor Padel Play Figure 26: Indoor Padel Play

Regarding who will appear in the videos, I've concluded that we'll need at least two people: one for close-ups and another to participate in plays or to rotate so it's not monotonous. To that end, I've searched through various websites and professional padel clubs that would be willing to do this in exchange for payment, of course.



Getting into the videos themselves, the video format is crucial, so we'll record everything in a way that allows us to use the videos both horizontally and vertically. The videos need to be classified, so I think it could be something like this (a video can belong to several levels and must belong to one level or type of movement):

- **Level:** Beginner, Advanced, Amateur, Semi-professional, Professional and a special one called Sunday players or something like that which would be focused on people who only do it for fun with their friends.
- **Movement / Other:** Serve, Stroke, Return, Positioning, Strategy Tips / Advice (this would be focused on beginners and would cover topics outside of the game such as the racket, clothing, footwear, etc.). Some movements would be grouped by level as well, since, for example, some volleys are more difficult than others...:

- Basic forehand and backhand.

- Simple serve.

- Volleys and trays.

- Doubles strategy.

- Reading the opponent's movements.

- Coordination and precision.

- Defensive movements.

- Serves with spin.



- DAY 3

1. Player Performance Metrics and Tracking

Progress will be measured using a system of clear and specific metrics, which will be presented to the user in a graphical and evolutionary way:

Measurement Area	Specific Metric	Aim
Striking Technique	Blade Angle, Point of Impact, Reaction time.	Diagnosing faults in the execution of the coup.
Track Movement	Speed, Area Coverage, Positioning. Displacement.	Evaluate efficiency and tactics
Precision	Percentage of blows of correct answers and errors.	Quantify the effectiveness of the blows made.
Consistency	Successful repetitions in a same exercise.	Measure the reliability of the blow under pressure or repetition.
General Progress	Week-by-week evolution (improvement graphs). overall performance over weather.	Visualize the trend and the Export to Spreadsheets



Figure 27: (DIA3) metric



2. Sketch improvement and ideas



Figure 28: (DAY3) ADVANCED ANALYSIS
Advanced video section

Premium Functionality Proposal: Guided Technical Analysis

This functionality focuses on correcting the

The form and progression of the training, and its high value will justify the subscription.

2.1. Video Player with Interactive Annotations

- **Video in Context:** The top part shows the replay of the shot that the application has identified as suboptimal or that you want to improve (for example, a failed backhand).
- **Feedback Labels or "Snacks":** These are superimposed directly on the player's image at the key moment of the stroke (impact, backswing or end of movement).
 - **Example of Etiquette:** A speech bubble pointing to the elbow might say: "**Elbow low: Insufficient arming.**" Another pointing to the foot position would say: "**Weight on heels: Transfers weight forward.**"
- **Pause and Deceleration:** The player would include advanced controls to view the play in slow motion or frame by frame, allowing the user to study the correction in detail.



2.2. Detailed Explanations of the Correction

- **Brief Diagnosis:** Below the video, a title and a concise explanation of the error detected by the analysis system would appear.

• *Example:* "Error detected: Low elbow and lack of torso rotation in the forehand punch."

- **Impact and Solution (The Why):** A pedagogical explanation is offered as to why that technique is incorrect and how it affects performance (e.g., "Low elbow reduces power and *topspin*,
Figure 29: (DAY3) Improve your smash by causing the ball to

MEJORA TU REMATE

POR QUÉ Y CÓMO CORREGIR

- Codo alto: +potencia, -lesiones.
- Golpe activo: +control, +control, +profundidad

PROGRESIONES DE ENTRENAMIENTO

1. Somba sin bola.
2. Con globo facil,
3. En escalera.
4. Punto libre direrjido.

stay in the net").

- **Reference Visualization:** This could include a small animation or image of a professional performing the movement correctly as a model.

2.3. Recommended Training Progressions

- **Practical Action Plan:** Specific exercises are provided to correct the detected defect. These exercises would be classified by difficulty.

- **Stage-by-Stage Progression:**

1. **Basic Exercise (Solo):** *Example:* "Practice the 'shadow' (without a ball) in front of a mirror, concentrating on keeping your elbow high."

2. **Precision Drill (Assisted):** *Example:* "Ask a partner to throw the ball very gently so you can focus only on the position of your elbow at impact."

3. **Resistance Exercise (Match):** *Figure 30: (DAY 3)*
execute only the corrected shot for 10 minutes of rallying."

PROGRESIONES DE ENTRENAMIENTO

1. Somba sin bola.
2. Con globo facil,
3. En escalera.
4. Punto libre direrjido.

Improve your smash Example: "Try to



3. Player tracking system

To analyze the user's padel matches, including their overall metrics such as reaction time, serves, positioning, etc., we will use a Google model (real-time pose estimation) that detects the person's pose and movements in real time. This model is called **MediaPipe-Pose-Estimation**. Its functionality will be implemented in our application as follows:

3.1 Track Measurements

Basically, when scanning, the player tracking system will take into account the predetermined **dimensions of the padel courts** ; these are **always the same** in all padel courts (**10 x 20 meters**) and other predetermined line dimensions as well.

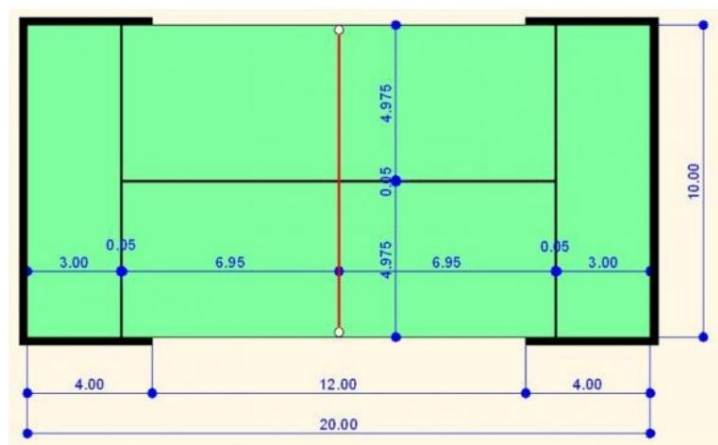


Figure 31: (DIA3)Track and measurements

3.2 Body tracking

Next, it will identify where the player's body is located and assign **keypoints** (key points, which are points that the model places on the different limbs of the body to create a virtual skeleton in which all the movements, speed, and position of the user's body can be seen at all times).

Specifically, **MediaPipe** uses 33 reference points to recreate the player's **virtual body** . Then, using AI, it detects where the player has fallen short and areas for improvement, linking these errors and areas for improvement to the application's educational materials.

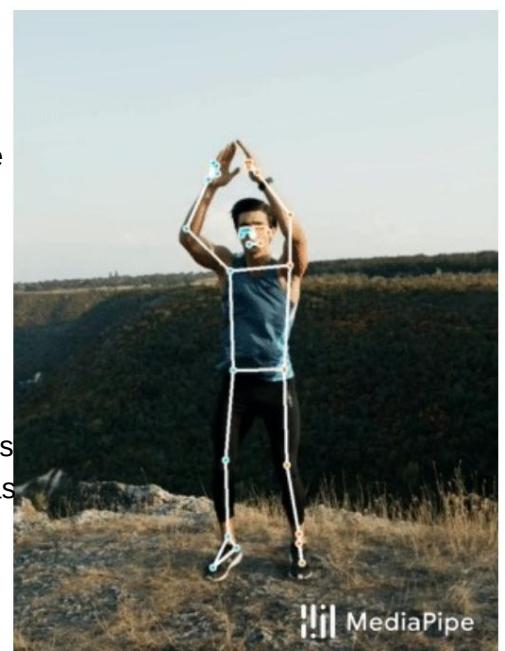


Figure 32: (DIA3)Track and measurements



3.2 Ball tracking

After detecting the user, it **would track the ball's position**, monitoring its location at all times, to internally map its trajectory. And, for example, it would know how quickly the player reacts, serves, receives the ball, etc.

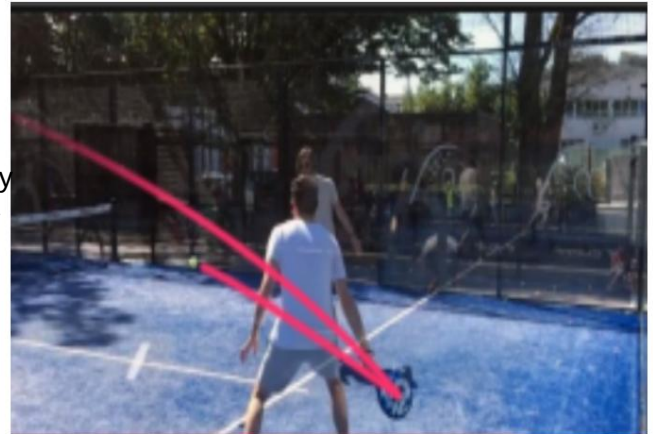


Figure 33: (DIA3)Track and measurements

3.3 Full tracking (court, player and ball)

As a final result, after the application's default settings, the field dimensions, automatically created using **Google's MediaPipe** AI model to generate the player's **virtual body**, and the **ball's** location, the AI would track the match as follows.

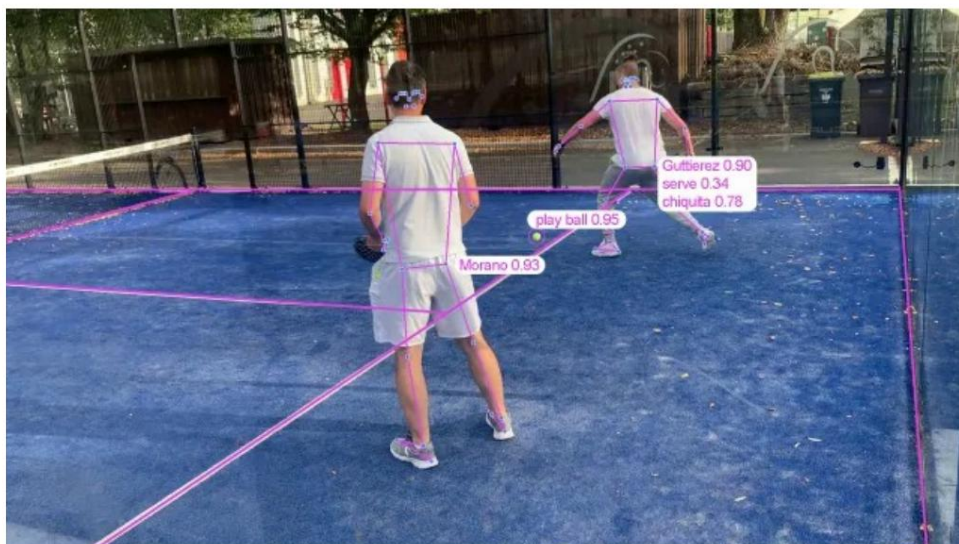


Figure 34: (DIA3)Track and measurements



4. INTERACTION PROTOTYPE



Figure 35: (DIA3) Interaction prototype

App name:
“Padel Analyzer Pro”

Main action button: Large red button with camera icon **“Start new recording”** ÿ is the user's key starting point (record your game/training).

Preliminary analysis section:

“My Analyses” ÿ Saved session library.

It allows the user to view their past results again.

Performance metrics (example in prototype):

Bar chart (probably errors/successes or reaction times).

Court heatmap ÿ shows most frequent playing areas (movement ranking).

Bottom navigation menu:

Home (main screen).

Statistics (full reports, trends).

Profile (player data, settings).



5. Recording videos from the application

On the first day of recording, we will focus on the videos for the application, which will teach how to perform a certain movement or play. For this, we will use the list we made the previous day and the classification, making recording easier since we can follow an order.

We will need light sources to generate more attractive videos with better visual results, different cameras and tripods for them.

Filming will take place throughout the day, starting at dawn, to capture better shots for the app's homepage, etc. We'll take two breaks of a couple of hours each to rest and recharge cameras and other equipment. We'll alternate filming between indoors and outdoors.

The videos have an educational purpose, so we are not so much looking for them to look beautiful or impressive, but rather for the movement being performed to be seen and understood. In addition, it will be recorded from different perspectives and also, if necessary, in slow motion to ensure good shots that allow the user to understand the techniques or movements.

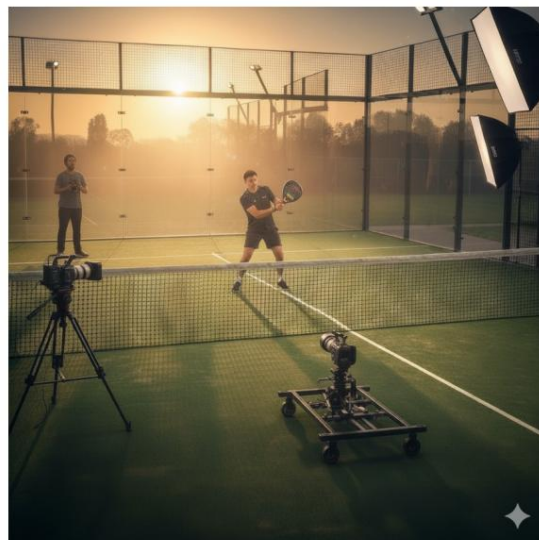


Figure 36: Example of recording.

We will also record some videos that will allow us to use them to edit videos and manipulate them as we wish, combining them with the rest to create videos without needing a physical person or space in the future, or at least being able to do it with another person but using their image since the users will already be familiar with that person and it will be easier for them to connect.

What I'm saying now would be using AI, but in the future, if more videos and content were needed, it could even help create content for social media. This way, people would always see the same person, even if the person managing the social media changed, so they would associate that person with the brand, which would never change and help create and maintain a brand image.



- DAY4

1. FUTURE IDEAS

User Interface (UI/UX) Design and Graphic Aesthetics

Navigation Structure (Screen Flow)

1. **Home/Dashboard:** Quick access to the main functions (**Record** and **Statistics**). It should show a summary of recent progress.
2. **Recording (Critical Flashback):** Screen focused on ease of use, including a **guide Visual camera placement** to ensure the quality of the analysis.
3. **Analysis/Feedback:** The central hub where the user sees their **detailed statistics**, **performance graphs**, detection of their **technical errors**, and **improvement tips** suggested by AI.
4. **User Profile:** Contains the user's **personalized goals** and complete history of recordings and all the **graphs and statistics** of its evolution.

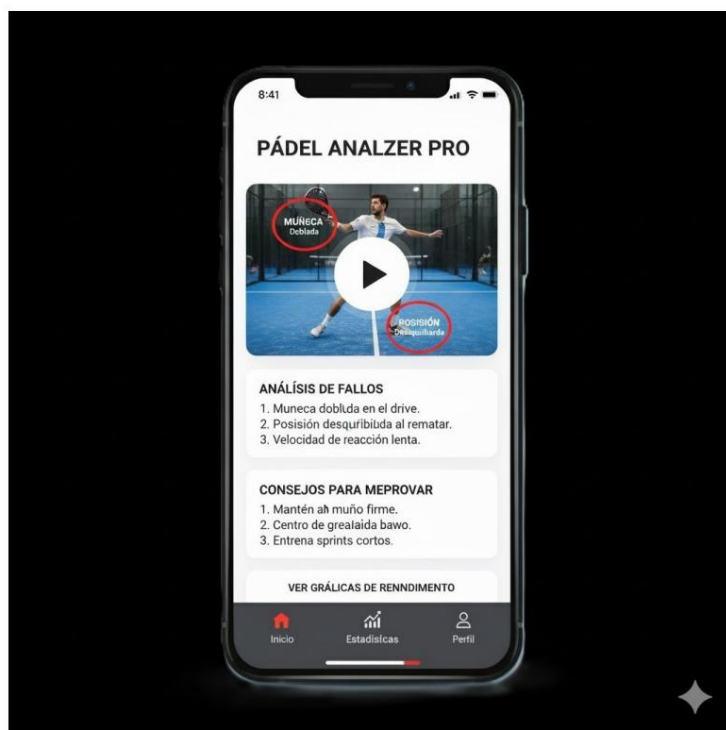


Figure 37: (DIA4)New designs

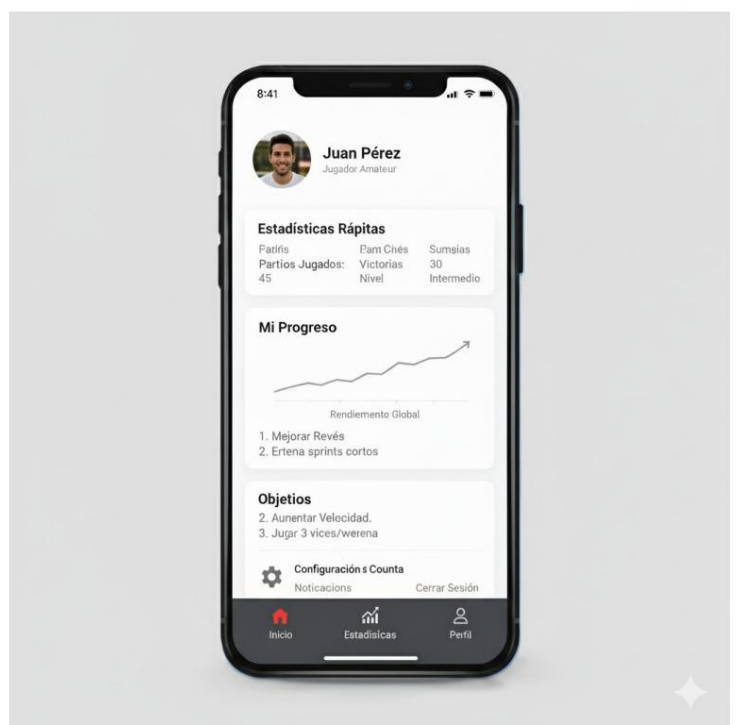


Figure 38: (DIA4)New designs



2. Justification for the Multi-Sport Expansion/Future Ideas

Applying video analysis technology (computer vision and AI) to other sports similar to padel is very efficient because many share the same analysis principles.

2.1. Reuse of Technology and Similar Analyses

The biggest advantage is that the application's core technology can be **reused** with minimal adjustments:



Figure 39: (DAY4) MORE SPORTS

- **Racket Sports (Tennis, Squash, Table Tennis):** The

analysis engine already knows how to detect a racket, a ball, and human movements. The key metrics are almost identical:

- **Impact Areas (Paddle/Racket):** Analyze the sweet spot of the shot.

- **Ball Speed and Reaction:** Essential in all of them, especially in Ping Pong.

- **Tactical Positioning:** Where to moves the player on the track (similar to a heat map).

- **Field/Net Sports (Soccer, Volleyball):**

Although they are different, the app can offer basic analysis functionalities

motion:

- **Soccer:** Analysis of stride, running speed, and positioning in the field.

- **Volleyball:** Height and angle of jump for spike or block, and positioning in defense.

- **Valencian Pilota:** Analysis of the striking gesture, force, and rebound accuracy.



2.2. Greater Segmentation and Market Niches

By offering analysis for a niche like **Valencian Pilota**, the application becomes instantly becoming the **market leader** in that area, attracting a very dedicated community.

2.3. Scalability and Return on Investment

Developing a "universal" sports analytics application is much more profitable in the long run. which would require creating a new app for each sport. Once the cost of the AI for a In sports, the cost of adapting it to another sport is significantly lower.

Example in volleyball



Figure 40: (DAY4) VOLLEYBALL EXAMPLE



3. Player tracking system

The MediaPipe model will primarily obtain two types of information: input data and evaluation criteria for each play, shot, or action of the player.

3.1 Input Data

We could also perfectly call it game data, but it's the data we obtain from the player.

1. **Coordinates:** obtains the X, Y, Z positions of each joint, using the **keypoints**.

This helps us know where the player is on the court at any given moment.

2. **Distances:** This measures the **distances between each joint** to determine the player's overall position. For example, the distances from the elbow to the wrist or from the shoulder to the hip. This helps us measure the efficiency of the player's positions.

3. **Angles:** This provides the **angles of each of the player's joints**. This information is crucial for improving technique. It indicates whether the player's actions and movements are efficient based on the angles of their joints.

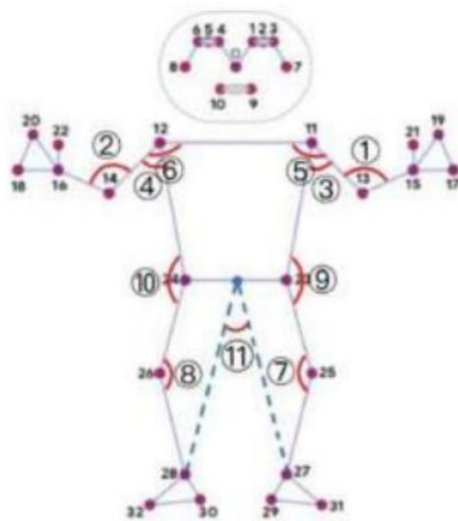


Figure 42: (DAY4)Measurements

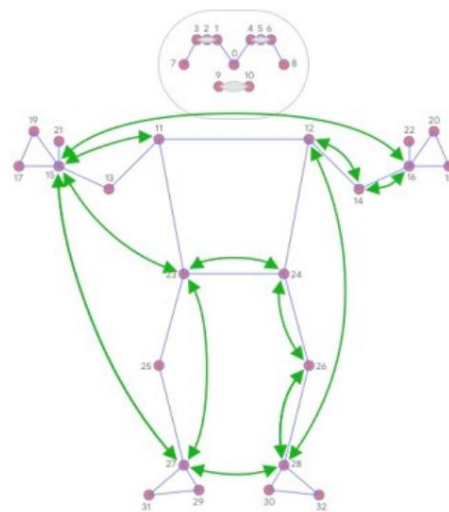


Figure 41: (DIA4)Measurements

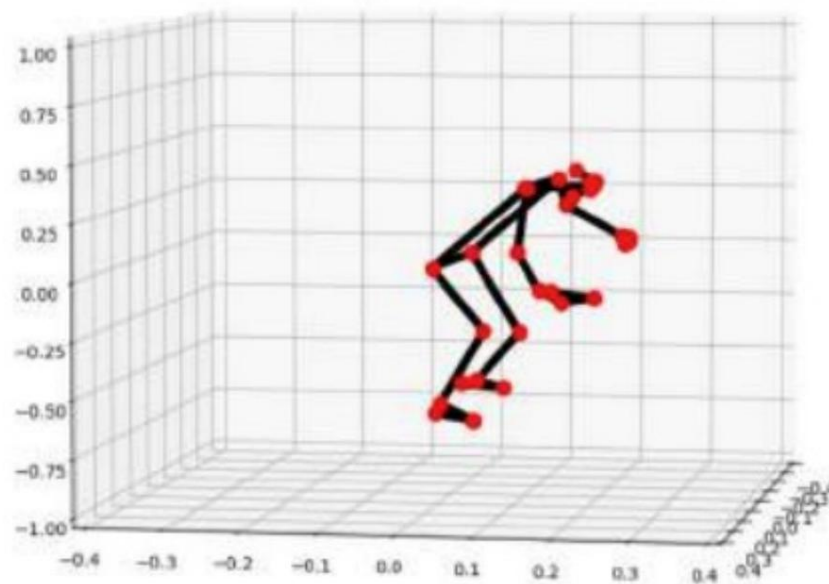
3.2 Evaluation criteria

The following are the criteria that MediaPipe will use to analyze each play and, therefore, the entire match of the application user:



1. **Execution technique:** utilizes the **distances and angles** between each **joint** of the player to analyze whether that **gesture is correct or not**. Also with the aim of **avoiding Injuries**: the user is informed of potential **overload risks** when using their current technique.
2. **Power and accuracy of the blow:** three main angles are measured (**torso-hip, elbow, shoulder**), the direction and control he has of the player's paddle at the moment of impact with the ball.
3. **Reaction and impact time:** analyze **by comparing the different positions** of the player between the **time** in which the **ball goes towards the player, starts and finishes the stroke**.
4. **Position and stability:** the **player's positions** are analyzed , especially when moving along the court, the **balance** he has, how far he **travels** and how **often**.

They also analyze, using **heat maps**, how **much of the court he occupies** during the match. The player, seeing if they are efficient or not. Detecting potential **weaknesses** in their gameplay. and leave areas uncovered.





4. Recording videos for (Social Media)

On this second day of recording, we can resolve any issues that may have arisen during the first day. Our day will be dedicated to recording content for social media, not just one-off pieces, but also videos and photos that can be uploaded periodically to keep things active. However, we will focus primarily on recording videos and deciding what kind of videos we want while we record the first ones.

As I recently mentioned, while recording the "filler" videos, we will research what videos we will make or how we want to make them and adapt them to our field. That's why we have a day dedicated to this, because we want to have a reserve of videos ready so we always have them available and can maintain high activity on social media, and even if we don't have to upload on a certain day, we'll have them there as a backup.

But how will we know which videos to upload and which videos to look for? Well, our idea is to create modern and easy-to-understand content so that everyone from the youngest to the oldest can watch, understand, and enjoy it. This doesn't mean all the videos will be the same, as videos that meet these criteria are very limited. Therefore, we'll aim to create videos geared towards children, teenagers, young adults, and finally, adults. We'll also keep in mind that within each of these categories, there can be different skill levels: professional, beginner, and even those who don't play padel. We won't focus solely on padel content itself, but also on padel facts, things you might not know, and so on. We won't just create videos on how to do a move or things like that, so we can attract more people not only to the app but also to the sport itself.

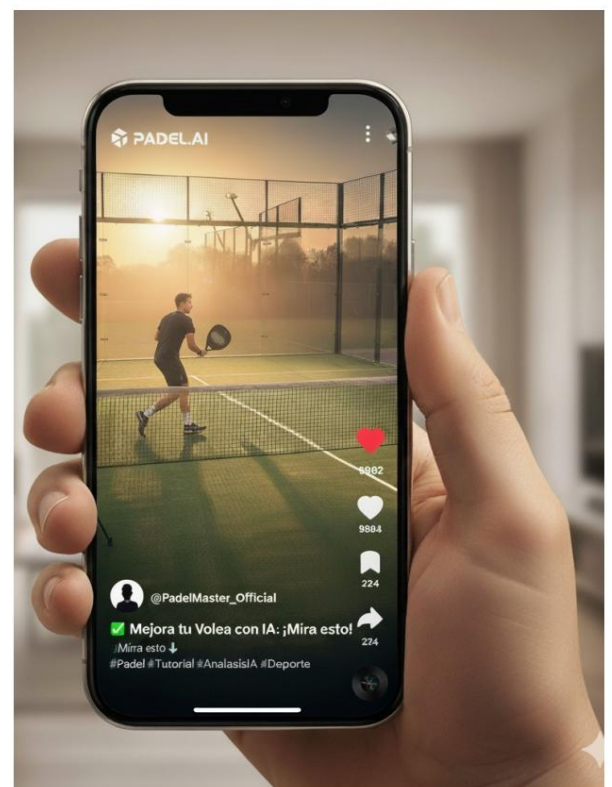


Figure 43: Social network format



- DAY 5

1. Improvements to the tracking system

There are several improvements to the application's tracking system that we can consider for the future.

- *Possibility of tracking a pair of players*

Since padel is very commonly played in pairs, the possibility of scanning a pair of two players. And apart from analyzing them individually so that each one can improve, also that they improve as a couple in the way they combine when playing.

The latter is very important for perhaps competitors or two people who usually play

They spend many hours together and need to know specifically what to improve on in the shortest possible time.



- *Instant tracking capability for tournaments*

For now, our application only tracks one player individually.

after the video recording is finished. That is, give the advice and provide the educational material.

after the game ended.

It would be interesting to have the option of being able to give short tips instantly to a player or to the pair of players (if we apply the previous improvement) to whom the follow-up.

This would be very interesting, to improve in small things, but at certain times

They are important and make a difference when competing. Thus, the coach could guide them

the player couple of. with the help too of the

Or simply for users who see it as an alternative or complement, to keep learning

also with small, instant tips.



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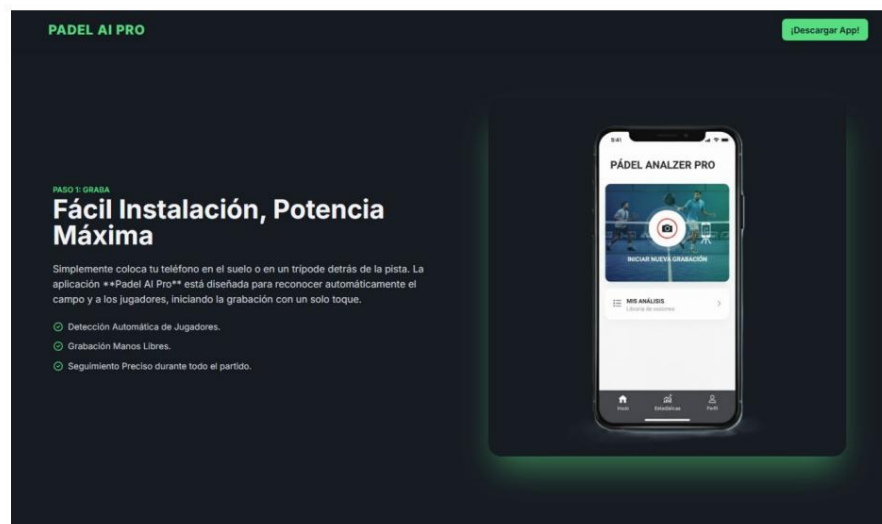
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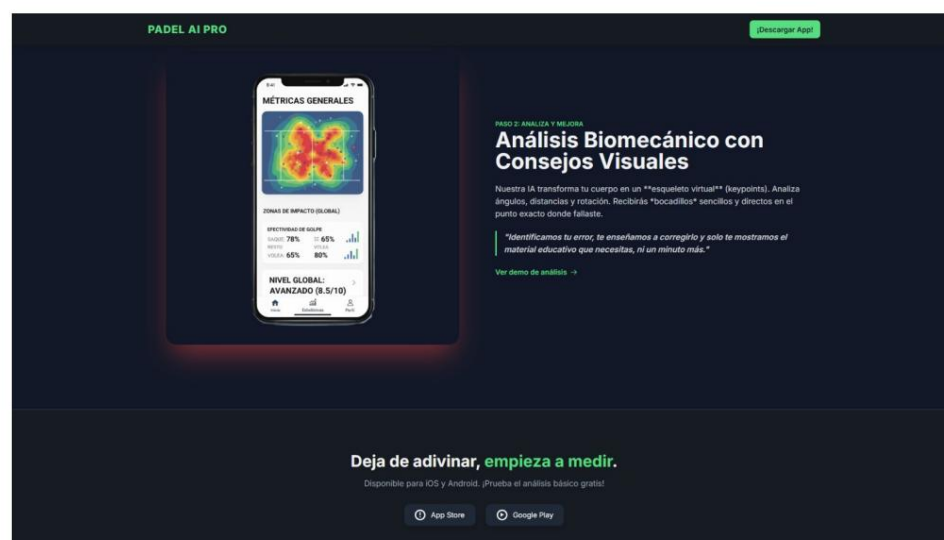
2. Initial website design

Next, we will design a website for our application. This website is primarily intended to promote the application to interested users, explaining and showcasing its innovative features.

In this initial design, they summarize the first options our application offers on the home screen, which relate to recordings. This highlights one of the central points of the application that tracks the player during the game.



This image shows an example of the application, in the metrics section that will analyze the player. Also, if the user is interested, we have included several buttons to download our application.





3. Video editing and integration

At this point in the project it's time to put the different parts together; here's how I add videos to the application.

The first step is to review footage from different cameras and determine which camera angles work best together to create a cohesive video montage. Following this, we'll begin editing, utilizing these various shots to clearly demonstrate the correct execution of each play and movement, highlighting key points and ensuring clarity. We'll edit all the videos to maintain a consistent aesthetic and style, so that the source is immediately recognizable. This will make them easily distinguishable, allowing other brands or individuals to instantly recognize the source even if it's not explicitly stated.

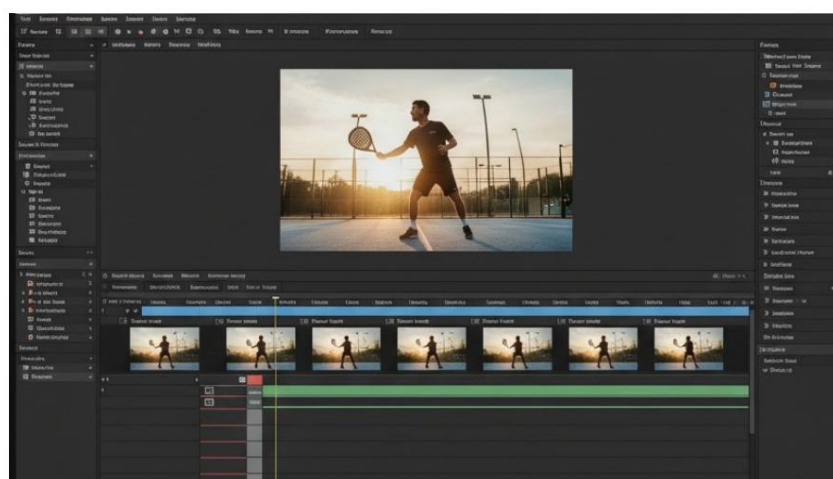


Figure 44: Editor

To integrate and insert the videos into the app based on the classification we previously established, and once all the videos are compiled, we will create a database to store the different videos and how they relate to each other. For example, different defensive or offensive techniques may belong to different levels of play. This will help the AI easily select the appropriate video for each situation.

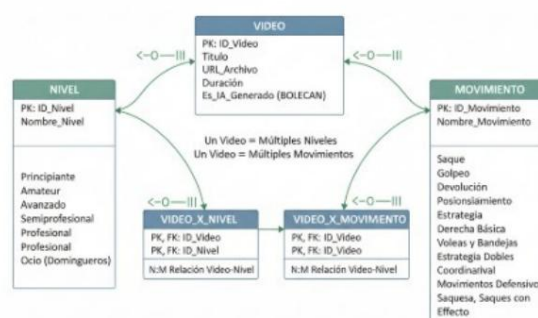


Figure 45: Database



4. External opinion

We chose 3-5 people to give us their opinion.

We'll let you explore the app for yourselves.

We don't usually explain too much, so we can see if they understand it without help.

We observe and note down doubts

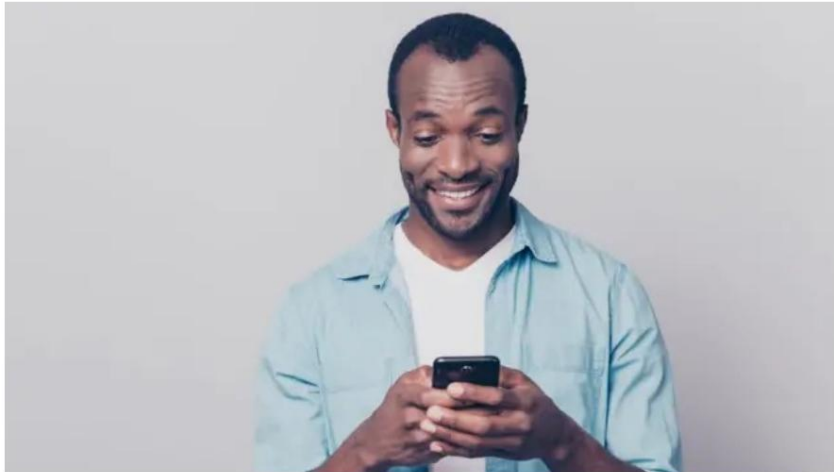
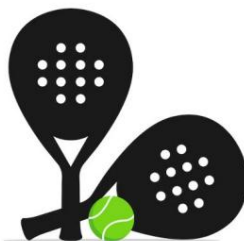


Figure 46: external opinion (DIA5)

5. Graphic Design Adjustments

- **Color Palette:** Use **sporty and dynamic tones**, such as **electric blue** for important details, on a base of **simple and clean** main colors (**black and white**) to ensure clarity.
- **Typography:** Use a **modern** font that ensures **maximum clarity** **reading** data and advice.
- **Iconography:** **Minimalist** and **highly recognizable** (e.g., padel racket, camera, graphic, trophy).
- **Fluidity:** It is crucial to ensure a **quick and smooth transition** between the **Record** and **Analyze** screens .

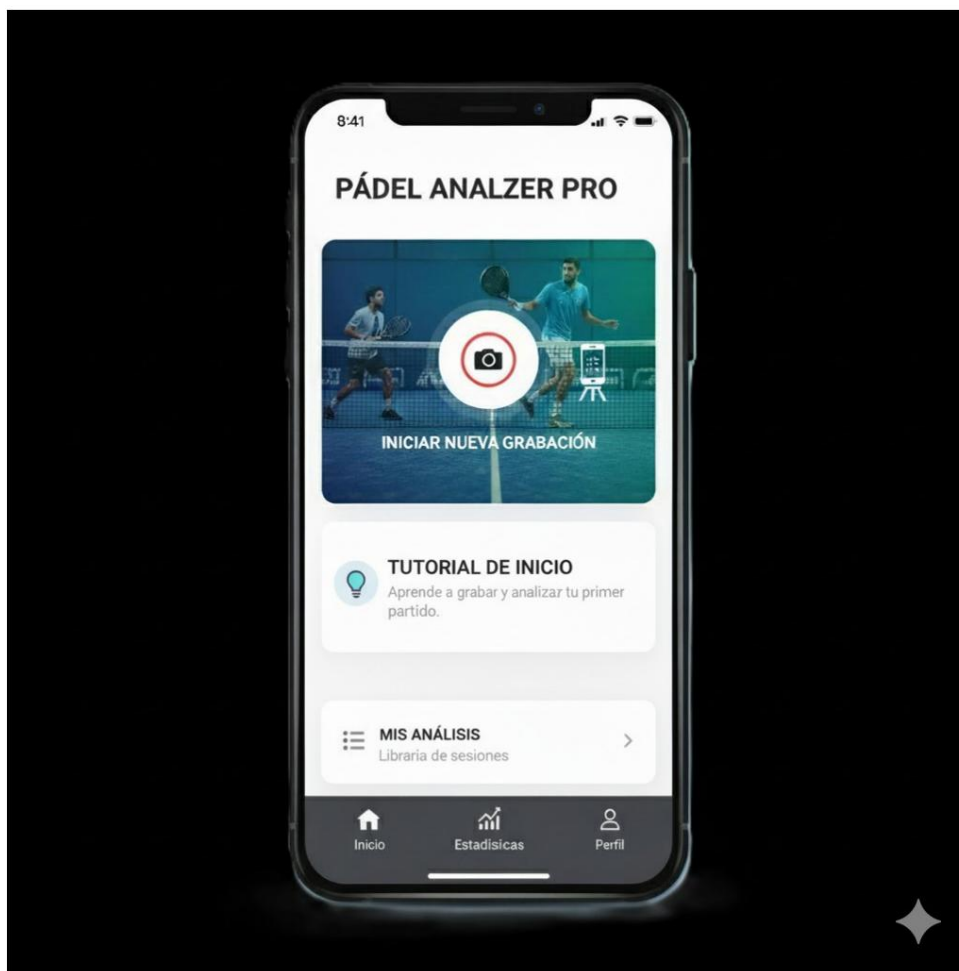




-DAY 6

1. Final Interface and User Experience (UX) Improvements

1. **Load Optimization:** Prioritize loading and immediate analysis of the videos for a real-time experience.
2. **Interactive Tutorials:** Implement step-by-step guides for new users, with a special focus on **how to position the camera** for optimal video analysis.



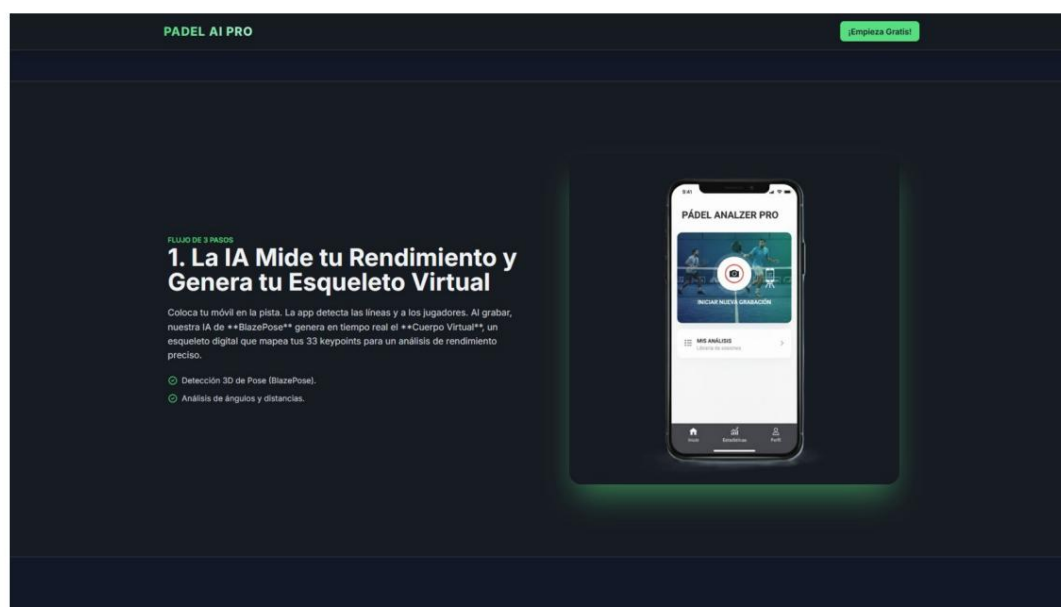


2. Initial website design

We've added a prominent title to the website, emphasizing the analysis of user plays and its accessibility to everyone. Then, three short notes reiterate the play analysis, the lack of additional time or financial costs, and the personalized educational support available to the user.



Subsequently, we changed the titles, commenting on the two main screens of the app, to make it clearer how we implement AI for player tracking during the game and subsequent analysis.



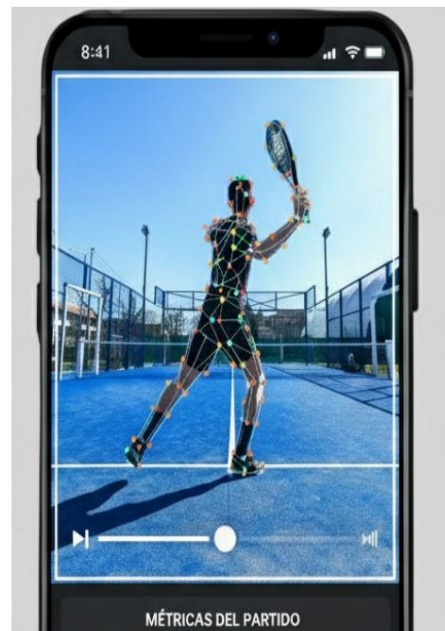


3. Monitoring documentation

Now we will need to document with a real example that the monitoring program will do.

Assigning keypoints to the player.

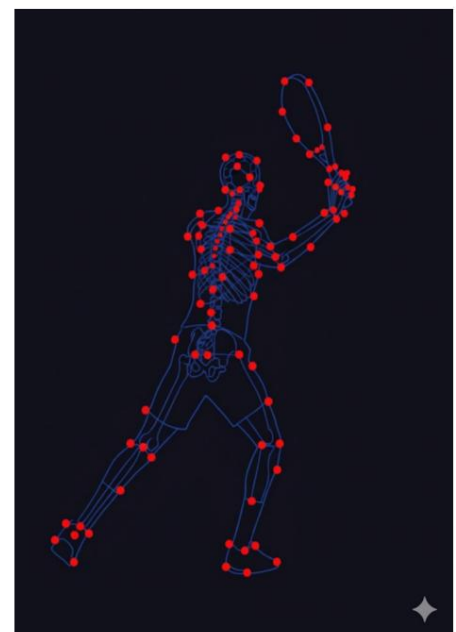
First, the tracking system will take the player's recording; here we have the example we will follow to document a practical example.



Creation of the virtual body

After assigning the **keypoints**, the tracking system will create a virtual model of the player's body. This will allow us to know the player's exact position.

Necessary to calculate the rest of the necessary information about the player: the distance between keypoints, limb angles.





4. Plan social media posts



CHANNELS: Instagram / TikTok > for short videos showing plays, tips and previews

Week 1 :

Monday: Teaser (simple image + phrase).

Wednesday: Basic explanation (mockup with text).

Friday: Benefit .

Week 2:

Monday: Short video showing prototype in action.

Wednesday: Padel tip + link to your page.

Friday: Invitation to participate as a tester.

5. Social networks

On the penultimate day, we edit the videos we recorded on day 4 for social media, create accounts on the different social networks where we will have a presence, and schedule the upload of some videos so that when we launch the application that day, they will be uploaded to all our social networks.

Firstly, we need to adapt the content a bit more, since for Facebook we will mainly use a more serious tone or an approach for older people, while on Instagram and TikTok we will focus on young people, since they look for things that are more stimulating and, in most cases, faster-paced. But this doesn't mean that we will only focus on that type of content, since we will also upload Facebook content and make videos with an explanatory tone, teaching things or interesting facts so that even people who don't play padel can become interested in our content.

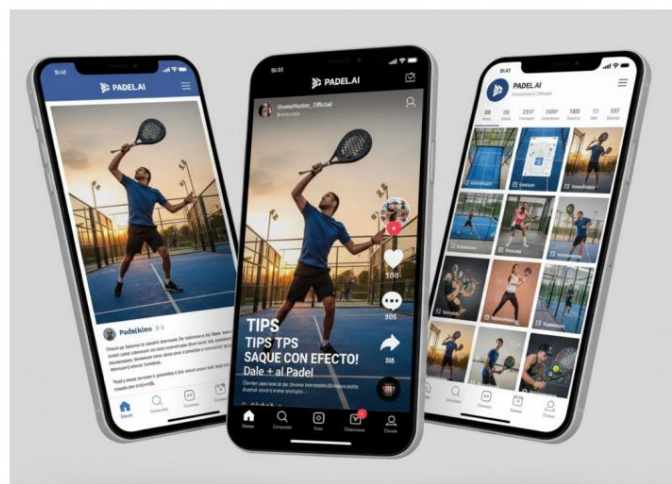


Figure 47: Social media preview



Once we have the videos created, we will create the social media accounts. We will make profiles for Facebook, Instagram, and TikTok, as we discussed earlier. They will all follow the same aesthetic, colors, etc. Now I will explain how we will approach each one:

- **Facebook:** we will aim for a more serious image, as mentioned previously, and showcase our product more, since it is a platform where we are interested in promoting our product because the audience there is very likely to consume it.
- **TikTok:** Our content here will be very dynamic and varied, not focusing so much on the product, although it will also have a presence, but it will have the same prominence as the rest of the content, seeking to grow in numbers in order to reach more people, creating a community and thus the more exposed we are, the easier it will be for our product to reach the audience we want, so by making videos that not only appeal to padel players we will achieve that.
- **Instagram:** it has to be a mix between TikTok and Facebook, for example, in reels the content will be the same as in TikTok, but then you will also have to upload stories to always be active, whether it's about the rest of the content or about padel or our app.
Now focusing on the aesthetics of our profile, since this is a very key point in a professional profile, we will seek to make our feed look nice, organized and easy to read, and allow people to see who we are or what we offer by pinning several posts at the top with our name or the app's name.

Across all our social media platforms, we will create automations so that videos are uploaded automatically, customers are answered automatically using AI, etc. We would possibly do this using N8N or a similar application.



- DAY 7

1. Presentation and Marketing Strategy

Key Message: "Your own digital padel coach in your pocket."

Presentation Demo:

1. **Practical Demonstration:** Show the **speed** of recording and **instant analysis**.
2. **Problem and Solution:** Clearly explain the problem (the lack of technical feedback). accessible in padel) and how the application solves it.

Commercial Focus:

- **Business Model:** *Freemium* with the possibility of **Premium subscriptions**.
- **Premium Value:** Access to **deeper analysis**, **advanced AI** advice, and possibly exclusive metrics or personalized training challenges.

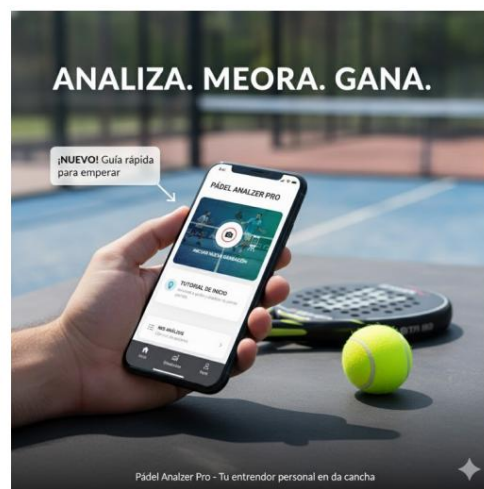


Figure 48: Commercial Strategy (DIA7)

2. Final website tests

Now I'll click the buttons to download the application for Android or iOS from the Google Play Store or the Apple App Store, depending on the operating system our client uses.



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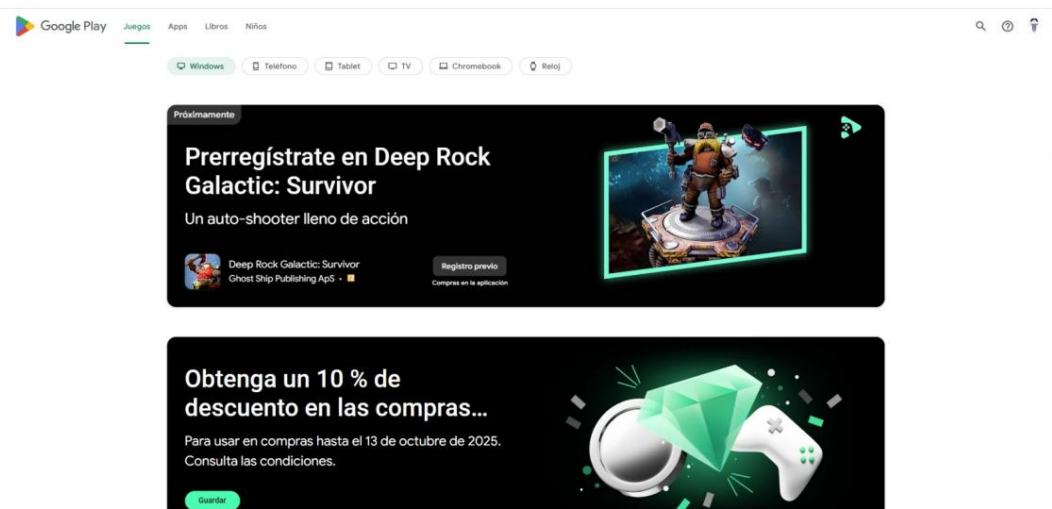
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This is important because other companies only focus on one operating system.

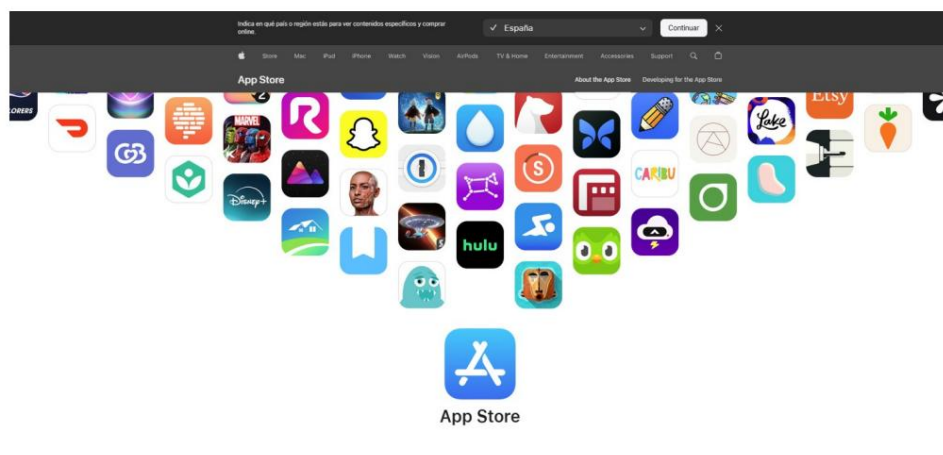
This is the fundamental part of the website: to get visitors to download the application.



Accessing the Play Store perfectly from the website.



And then we were able to access the App Store perfectly from the button on the website.





3. Monitoring documentation

2.1 Calculation and assignment of distances between keypoints.

It will then calculate all the distances between limbs and display the angles between them, using the previously created virtual body.

This will help us calculate the player's moves.

Obviously, for this the program will have performed the previous steps, with each of the player's movements.

2.2 Play detection

Then it will analyze all the information on angles and distances between keypoints of the virtual body, throughout the entire match. Following the evaluation criteria, it will provide small tips, which will be shown to the end user in the application.

Then it will transfer them from the virtual body, to the mapped image, and finally to the frame of the real video.

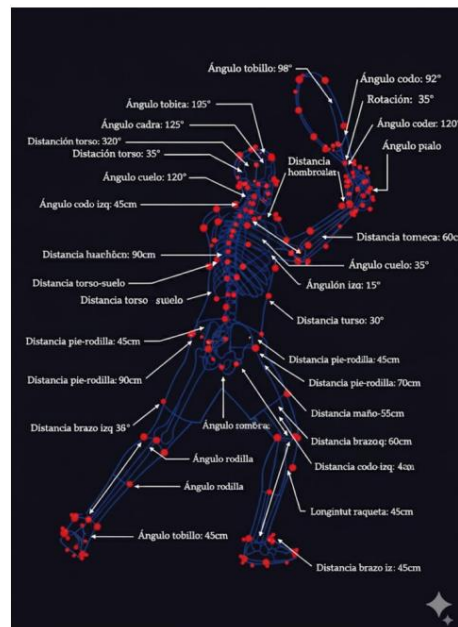


Figure 49: (day 7) follow-up

4. Promotional Event On the

last day of work, we will organize an event for the launch of the application. This will make it easier to raise awareness, even if only locally. This will also help us create content for social media and open doors to holding similar events in other locations if it is successful, or simply to gradually increase our visibility in different places.

The event will take place at [Pàdel Play Sant Vicent](#) And we'll rent everything out so that everyone can go and play for free and try out our app, so whoever goes will try out our app.

It will be all day long, aiming for maximum public attendance. On Saturday or Sunday, we'll have catering and plenty of free cold water for attendees, and we'll bring at least one food truck so people can eat there and make the most of their time. We'll also try to collaborate with Mano de Santo, an energy drink created here in Alicante. We'll also be raffling off padel equipment. We'll be recording everything to create content to upload to social media.

We will start promoting it a month in advance on social media, as we are looking for the highest attendance. We will involve customers by letting them choose, through voting, etc., the equipment to be raffled off, what they want the food truck to be, and other details of the event.



In addition to this, we would give away some merchandise to the first 50 attendees and the first 100 to register on the application or something like that, encouraging its use.



Figure 50: Event

5. Review of functionality and final presentation

We checked if everything was working properly.

Menus: All access points work. Screens: They look consistent (colors, typography, etc.).

Navigation: flows well from recording session

Communication review

Logo + name: finalize all the material

Social media: The posts accurately reflect the value proposition

Final presentation

Prepare a short step (2-3 minutes) that shows:

The problem (padel players want to improve their level)

The solution (your app that analyzes games using only your mobile phone)

A quick demo of the prototype



Gantt charts (in Jira)

To create the tasks for the five members of the project, we have created two Gantt charts, one divided into sections of the project and another according to the tasks of each person.

Gantt chart according to project sections

In the first diagram we have divided the project into five main sections:

- 1. Project definition and planning
- 2. App design and aesthetics
- 3. User experience and prototypes
- 4. Tracking system and technical functionalities
- 5. Multimedia content and learning
- 6. Review and final presentation

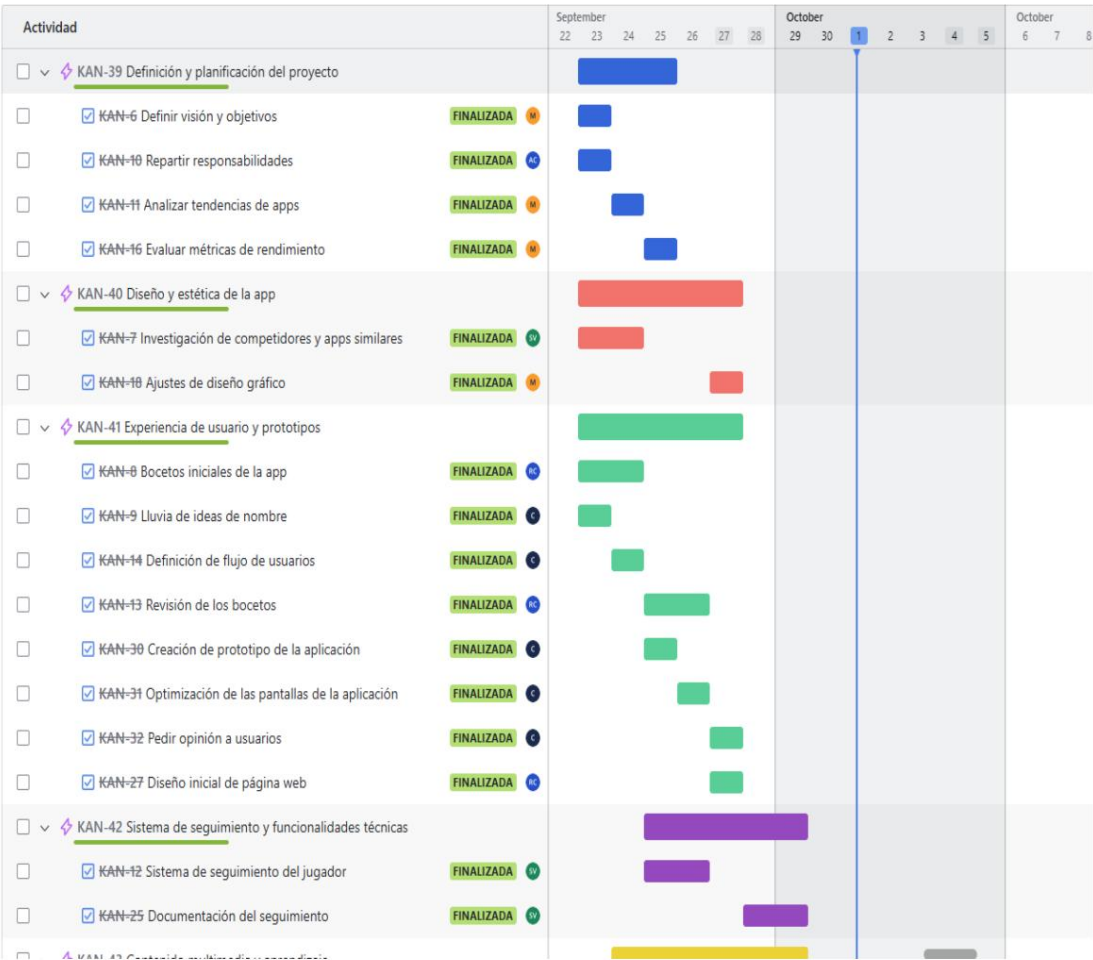


Figure 51: Jira diagrams

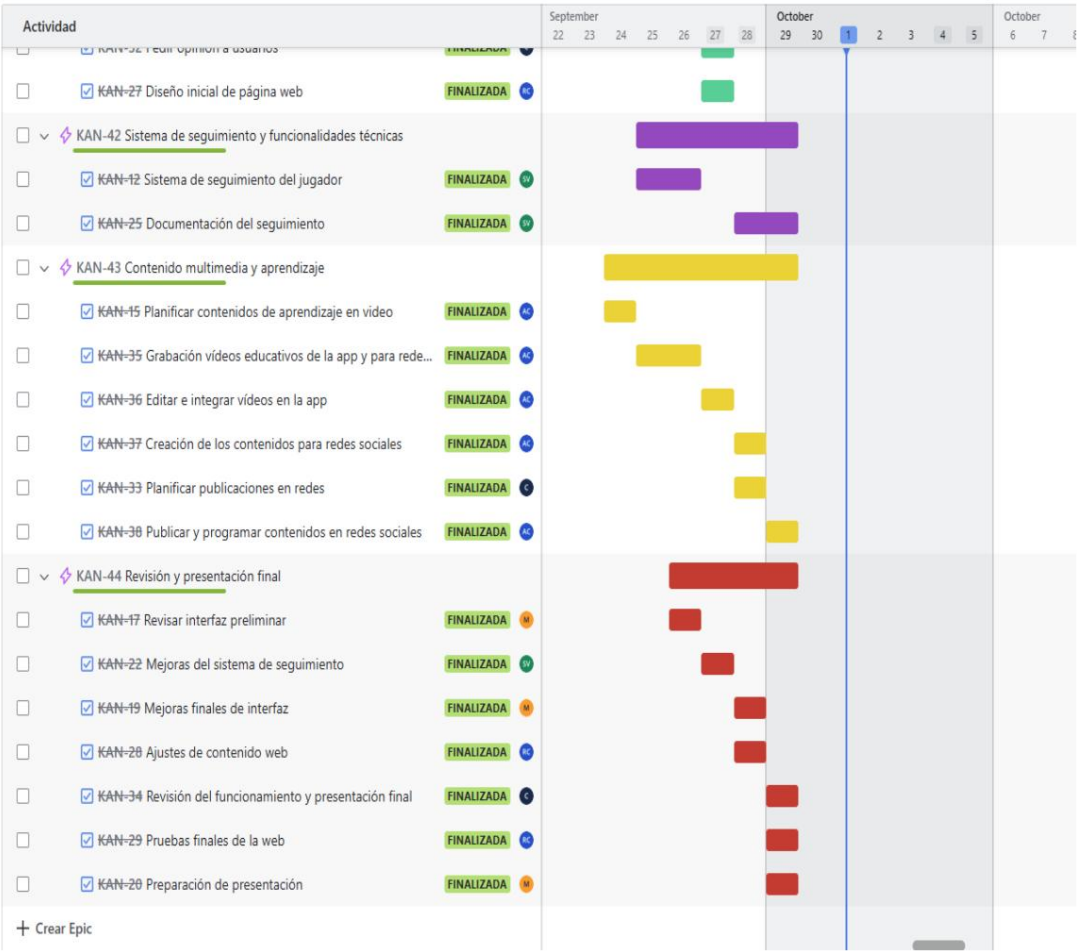


Figure 52: Jira diagrams



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8.2 Gantt Chart according to the project members

In this second diagram we have divided the project tasks by each member of the project.

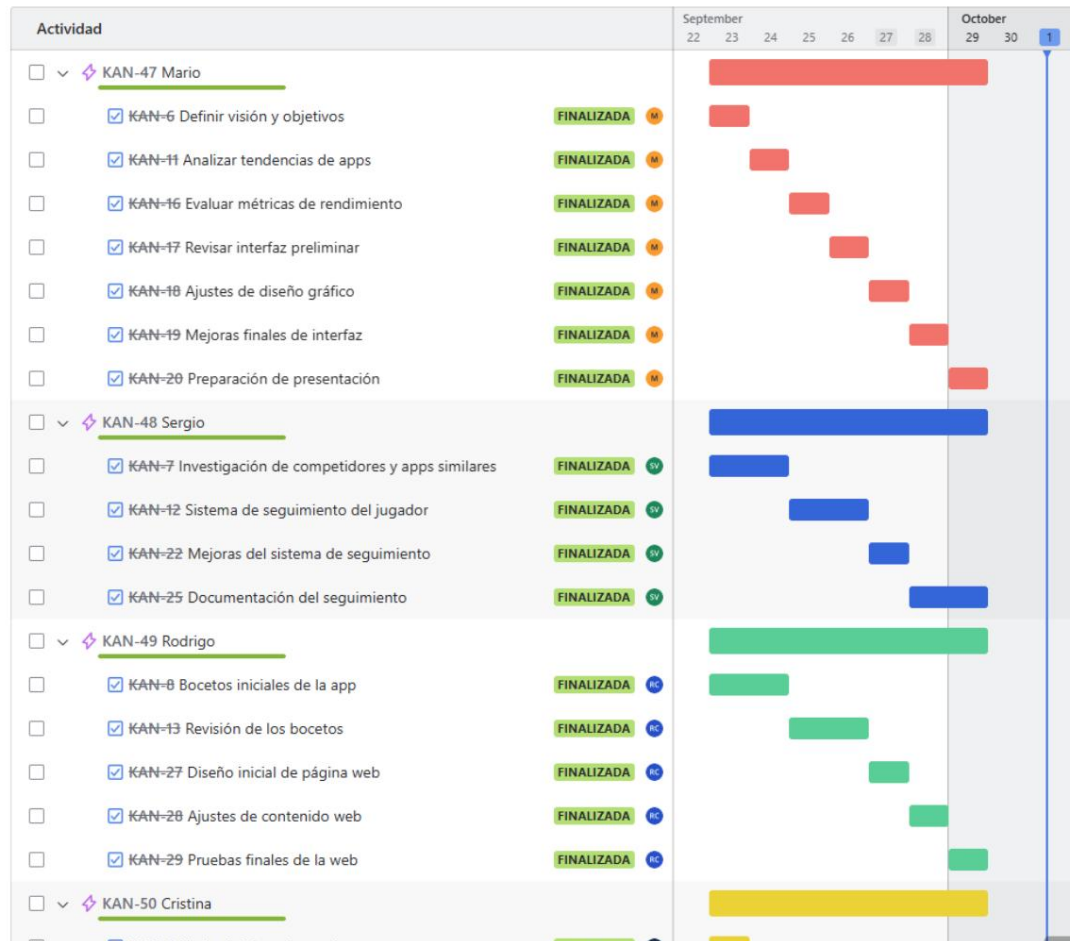


Figure 53: diagram

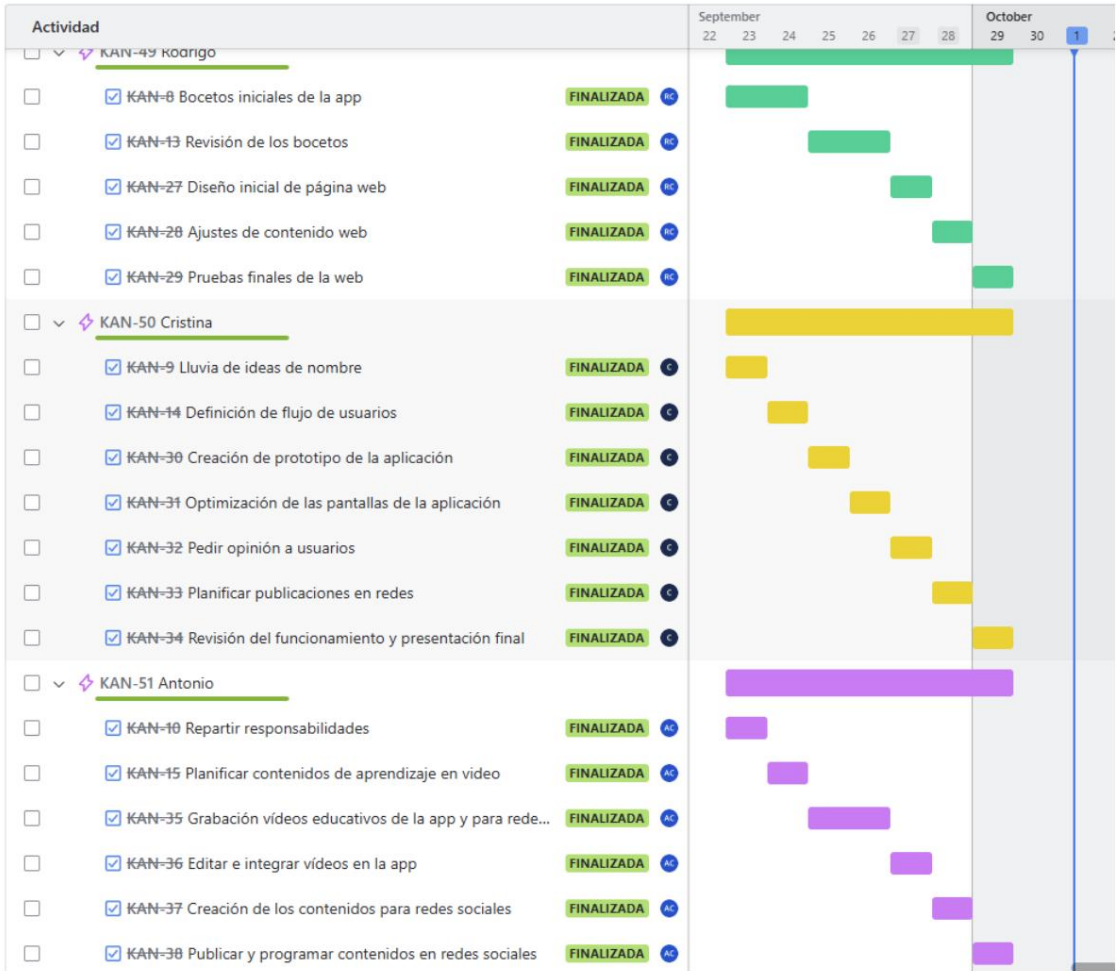


Figure 54: diagram



9. How to add a new team member and task to a project in Jira

9.1. Creating a Jira account and a new project.

First we will go to the Atlassian website ([Jira](https://www.atlassian.com)) and we will click on Login with the option we like best.

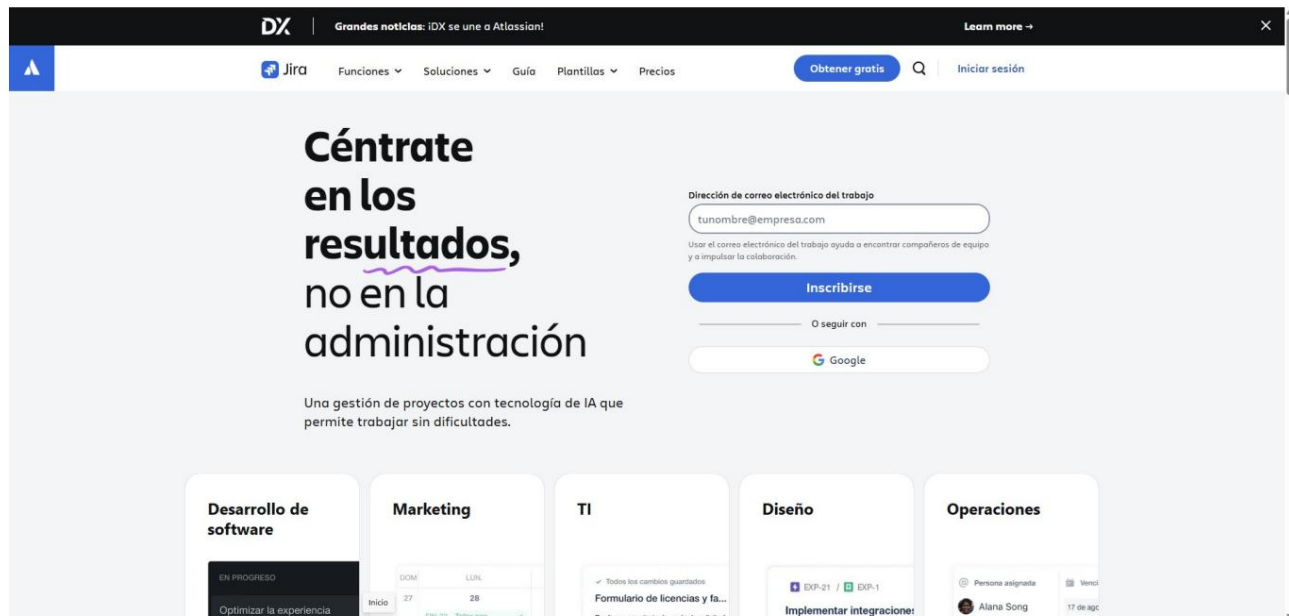


Figure 55: member

It will ask us for a name for our site on **Atlassian**.



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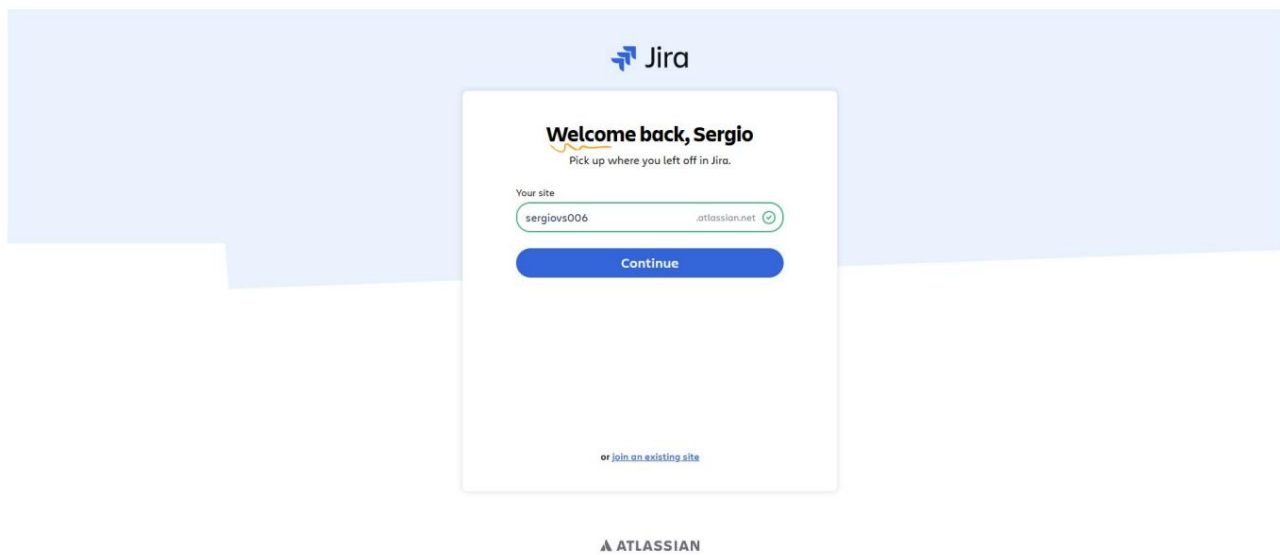


Figure 56: member



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Next, we will choose the project's working **methodology** .

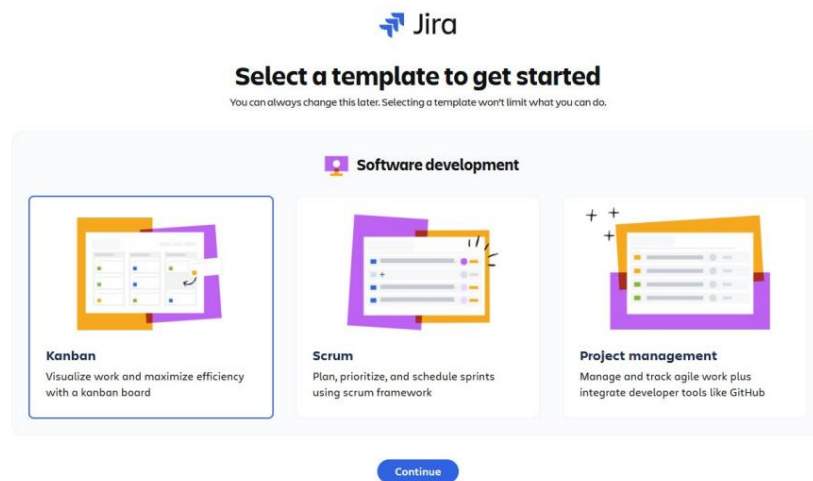


Figure 57: method

Now we'll give our project a **name** .

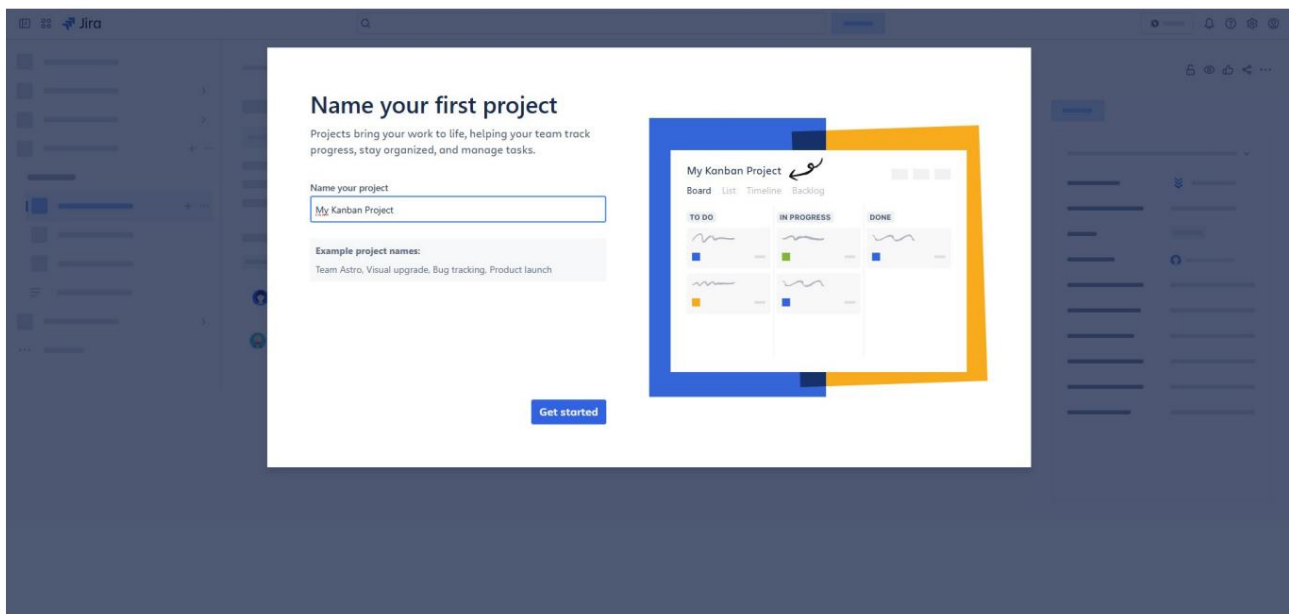


Figure 58: method



Next, we will need to choose the type of tasks and task statuses we need for our project. We can simply click continue, as we can change that information later.

And then our project would be available.

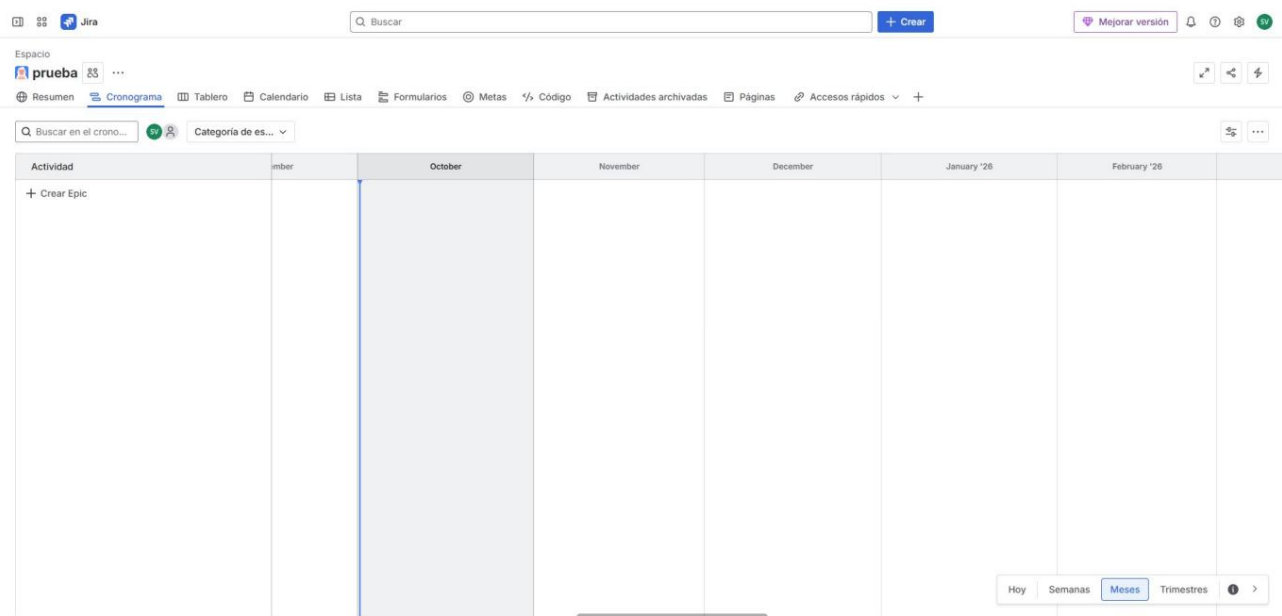


Figure 59: project

9.2 Adding a new task

We'll click on the three dots to the right of the project name.

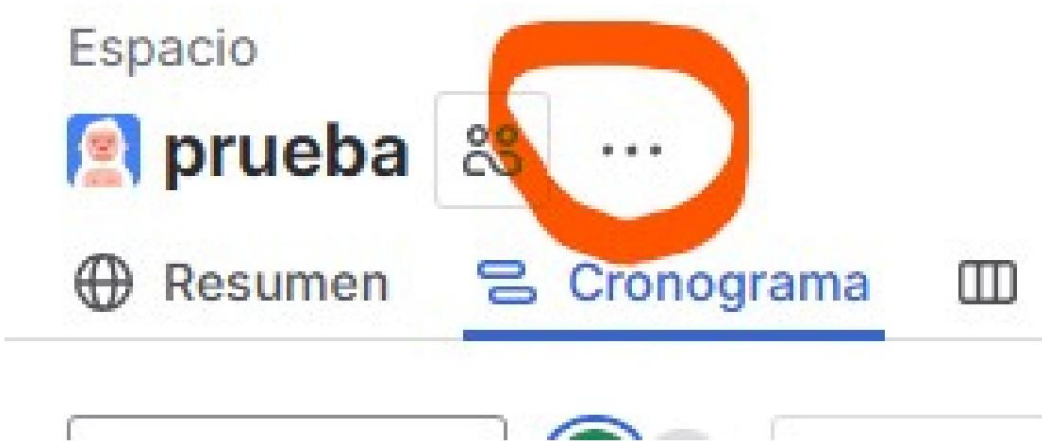


Figure 60: task



We'll click **Add people**.

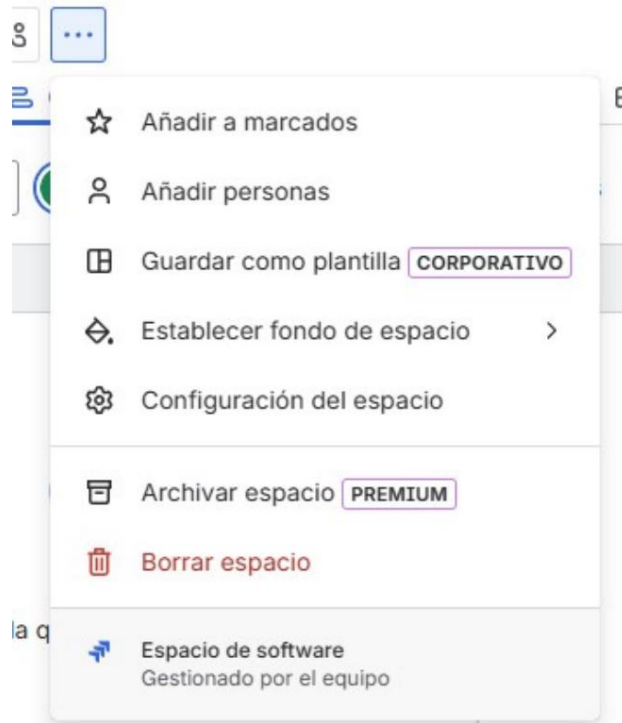


Figure 61: People. Now

we can enter the email address of the person we want to add to the project and also assign their role. And finally, click the Add button.

Añadir personas a prueba

Nombres o correos electrónicos *

por ejemplo, Cris, cris@empresa.com

+ criloppiq

+ mariodeustorios891

+ Antonio Penalva castillo

+ Rodrigo Viso Cain

o añadir desde

Google

Slack

Microsoft

Rol *

Administrador

Este sitio está protegido por reCAPTCHA y se aplican la [Política de privacidad](#) y las [Condiciones de servicio](#) de Google.

Cancelar **Añadir**

Figure 62: People



9.3 Create a new task in the project

We'll go to the Schedule section .

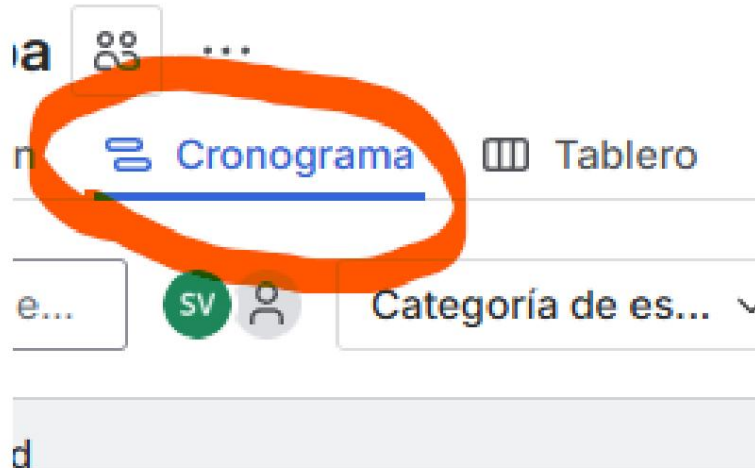


Figure 63: Task schedule

We'll add an Epic.

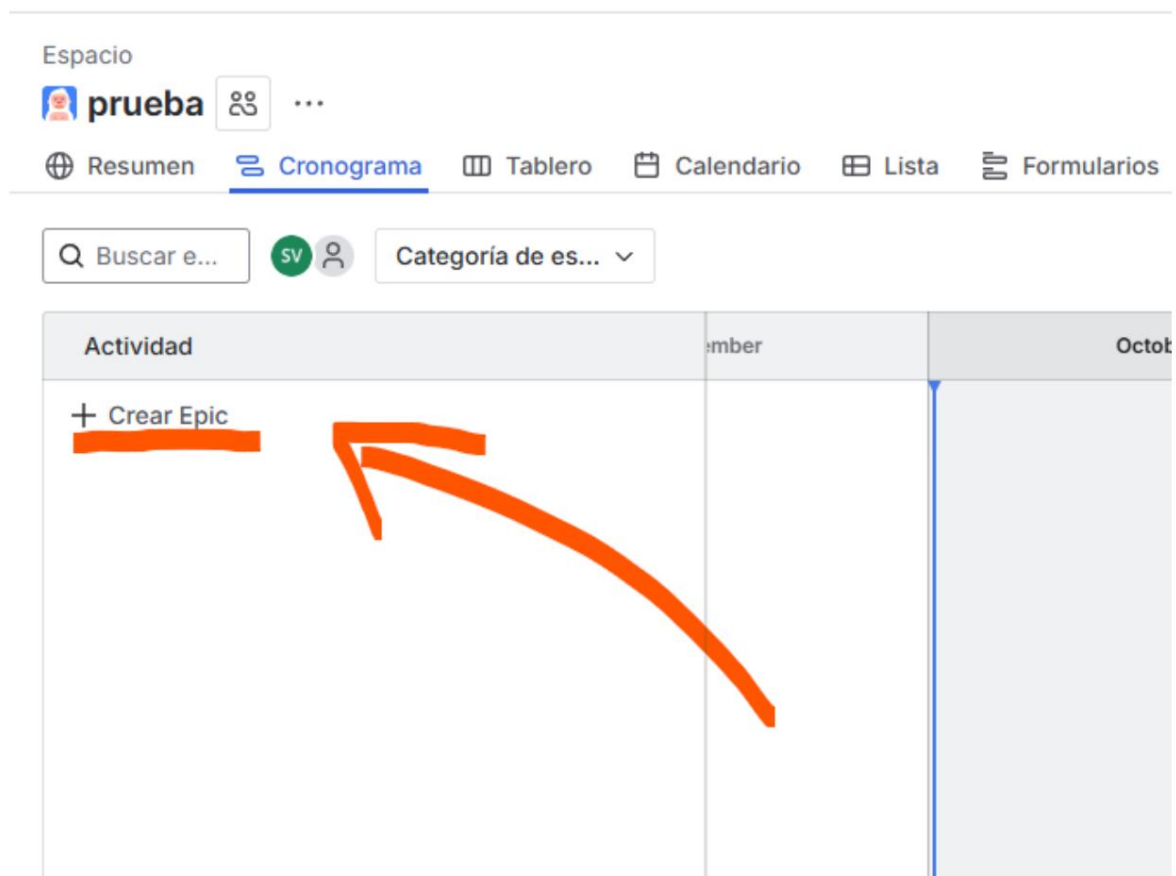


Figure 64: epic



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We'll give the **Epic** a name (it's used with a task container, it groups them together) and press **Enter**.

SV

We'll give it a plus sign so we can add a task.

SV

Actividad	Member
☐ PRUEB-2 Grupo de ejemplo	
Crear Epic	

Create child work item



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We'll name the **task** and press **Enter**.

Actividad	Member
☐ ⚡ PRUEB-2 Grupo de ejemplo	
☒ Tarea de prueba	

Once the task has been added, we'll click on it to change its settings and add all sorts of information, such as its status, the task owner, etc.

Actividad	Member
☐ ▾ ⚡ <u>PRUEB-2 Grupo de ejemplo</u>	
☐ ☒ PRUEB-3 Tarea de prueba	TAREAS PO...
+ Crear Epic	



Espacio

prueba

...

Resumen

Cronograma

Tablero

Calendario

Lista

Buscar e...

sv

Epic

Categoría de es...

Actividad

PRUEB-2 Grupo de ejemplo

☒ PRUEB-3 Tarea de prueba TAREAS PO...

+ Crear Epic

Hoy

Semanas

Meses

Trimestres

i

>

Actividad de Jira

PRUEB-2 / PRUEB-3

Tarea de prueba

Tareas por hacer

Descripción

Actividades vinculadas

Detalles

Persona asignada

Etiquetas

Principal

Fecha de vencimiento

Team

Start date



- Conclusion

The development of an advanced application that will merge **Artificial Intelligence** with sports analysis in padel **will demonstrate** that the planning and structure of the project **will be** as vital as the central idea.

The planning phase **will be** essential for setting goals (such as 80% AI accuracy and fast processing time) and identifying risks. Forecasting key resources (the AI Engineer and the Padel Expert/Coach) and managing the high costs of cloud infrastructure (GPUs) **will be** crucial decisions. This foresight **will ensure** that the project not only **gets off the ground**, but **does so** on a stable technical and financial footing.

The Project Closure phase **will ensure** long-term quality. Quality Verification **will require** rigorous cloud load testing and technical acceptance of the AI model by the Padel Expert, **confirming** the reliability of the final product.

The documentation, cloud architecture diagrams, and Operations Manuals **will become** the knowledge transfer. This **will ensure** that the application **can be maintained**, scaled, and evolved by future teams without relying on the original staff.

Finally, the Project Evaluation **will transform** the experience into knowledge. By documenting the **Lessons Learned** (such as the need to improve GPU cost estimation or accelerate data labeling), the team **will generate** protocols that **will improve** efficiency and **reduce** risks in future AI developments.

In summary, the success of this initiative **will depend** on a disciplined project management approach. and cyclical, which **will ensure** that a complex idea **is transformed** into a tangible, sustainable product and of high quality.



Webography

Project Management with AI	AI Project Management: Tools, Use Cases... (Triskell Software)
Methodology and Life Cycle	AI in software development (IBM)
Technological Risk Analysis	How to conduct a risk analysis and examples (Asana)
Padel: Biomechanics and Analysis	World Padel Tour - Analysis Articles Technician (Example)